cost will result from the unit cost by the quantity of access to be provided.

HISTORY OF CHANGES

PAGE NUMBER	CHANGES MADE
PART A	
Page 6	The "list of Deliverables" has been modified considering Project officer comments: *D1.4 Periodic report, period* 1; *D1.5 Periodic report, period 2*; *D1.6 Final report* and *D1.7 Final public report*, have been removed as they will be required automatically by the system.
Page 9	The beneficiary responsible of the Deliverable 8.2 IPR management has been changed, from LASEA to Ceit-IK4.
PART B	
Pages 25-26	An estimation of the total machine cost itemization has been included in section 2.2.1
Pages 28-29	An explanation of the expected performance enhancement for each one of the use cases has been included in section 2.1.1.
Pages 31-32	Additional information to the environmental and social impact has been included in section 2.1.2
Pages 39-41	A preliminary Business Plan for the three use cases has been included in section 2.2.2
Page 48	The following sentence: "*If that is not satisfactory, then the Project Board will make a decision and, if necessary, will ask for the authorization of the EC*" has been removed.
Page 48	The following sentence: *"Conflict Resolution – […] Procedures will be provided in the CA to be signed by the partners before the beginning of the project"* has been modified as follows: *"Conflict Resolution – […] Procedures will be provided in the CA to be signed by the partners before the signature of the grant"*
Page 51	The OTHER DIRECT COST TABLE of partner VISUM has been updated.
Page 52	The OTHER DIRECT COST TABLE of partner CIC has been updated.
Page 52	The OTHER DIRECT COST TABLE of partner RESCOLL has been updated.
Page 52	The OTHER DIRECT COST TABLE of Third Party Rescoll M. has been included.
Page 71	The description of RESCOLL TECHNOLOGY CENTRE OF MATERIALS has been changed.
Page 75	The following sentences has been included in section 4.2."*For all the subcontract activity foreseen in this project the identification of the respective subcontracts will be done on the basis of the best value offer. According to the GA regulation on sub-contracting the criteria used by the partners to assess the "best value" of the different potential subcontractors will be done and proposed on the base of: market tendering, partners' standard practices, commercial agreements already in place. Whatever will be the criteria used by the concerned partner for the respective subcontract, a clear justification of the "best value" will be provided to the Commission in advance of the subcontract finalisation*"
Page 79	The description of the activities subcontracted by RESCOLL has been changed.
Page 79-80	RESCOLL Manufacturing has been included as Third Party of RESCOLL
Page 75-80	The estimated amount corresponding to the tasks to be accomplished by third parties or subcontractors has been included in the corresponding table. Besides, the description of the Linked third parties has been moved and included in the table instead of below it.

Page 82	The following sentences has been removed: *We confirm that the proposal did not raise any ethical issue, so the table in the administrative form has been fulfilled in that way. The proposal has taken into account all ethics issues described below and none of them applies: 1. HUMAN EMBRYOS/FOETUSES, 2. HUMANS, 3. HUMAN CELLS / TISSUES; 4. PERSONAL DATA; 5. ANIMALS; 6. THIRD COUNTRIES; 7. ENVIRONMENT &HEALTH& SAFETY; 8. DUAL USE; 9. MISUSE; 10. OTHER ETHICS ISSUES.* The ethic issues mentioned in the Ethics Summary Report has been considered and a new deliverable D9.1 EPQ- Requirement No. 1 will be prepared and submitted in M6.
Page 83	An Appendix I (Appendix I. Figures from Work Packages description) has been added to include the figures related to Work Package description.

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5.2 SECURITY ..... **Error! Bookmark not defined.**
6. APPENDIX I - FIGURES FROM WORK PACKAGES DESCRIPTION ..... Error! Bookmark not defined.

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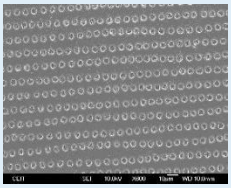
1. EXCELLENCE

Industrial markets (e.g. automotive, aerospace, biomedical or tooling) need to produce high added-value products with new and improved features (such as wear resistance, biocompatibility or optical properties) without increasing costs, rationalizing the use of resources and being produced under the minimum required production rate.

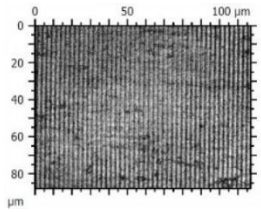
One of the methods currently used to enhance the product performance is the improvement of the material surface properties. **In this context, Laser Surface texturing,** which is the process of obtaining functional surfaces by applying a specific type of pattern or roughness onto the surface to change its properties using a laser technology, **offers a significant commercial potential, as it allows to improve product performances without altering the chemical composition of the surface or adding any coatings.**

The use of laser technologies allows surface texturing in the micro/nanometer scale, enabling the creation of tailored functional properties such as wetting, hydrophobicity, tribological characteristics of surfaces (i.e. friction, wear resistance, etc.), reflectivity, aesthetics or even anti-bacterial properties. Laser sources operating at pulse durations of nanoseconds (short) and picoseconds and femtoseconds (ultra-short) are exploited for this surface functionalisation.

Presently different laser texturing techniques such as Direct laser ablation (DLA) or direct laser interference patterning can provide a reproducible functionality. DLA consists in removing material from small features (spots of up to 2-200 □m with femtosecond lasers) in the target surface. However, this technique defines a spot with each pulse, leading to low processing speed[1]. DLIP can substantially increase process speed. This technique consists in the superposition of coherent light beams to form an interference light pattern composed by lines or dots periodically ordered. The area patterned with a single pulse covers the beam size (around 1cm^2), increasing processing speed substantially compared to DLA. However, processing materials with DLIP usually requires a laser source with a pulse duration in the order of the nanoseconds. Processing materials with such length of pulse tends to heat the surface of the material. This is particularly critical for metals, which have a higher thermal conductivity than other materials, limiting the minimum size of the achievable structures for metallic materials to a few micrometres.

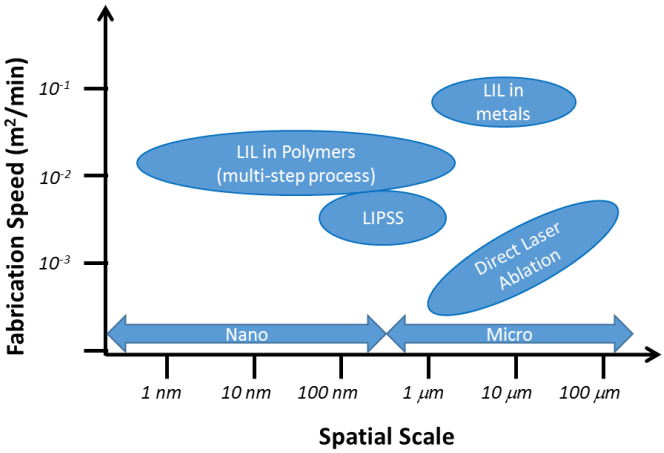
Technique	Feature Size	Main Advantages	Main Drawbacks	Pattern
DLA	2 – 200 □m	Features can be defined in precise locations of the component	Process speed is too low, it is a sequential technique.	 FIGURE 1: 2D Patterned grid by direct laser ablation (Source CEIT)

[1] HL Costa, IM Hutchings, Some innovative surface texturing techniques for tribological purposes, Proc IMechE Part J: J engineering tribology 2015, vol. 229(4) 429-448 [2] Laser4Fun Project-Newsletter-September-2016

DLIP	▪ 2 – 200 □m (metals) ▪ < 100 nm For metal the minimum size of achievable structures is limited to a few micrometres	Process speed is cost-effective ▪ Features cannot be defined in precise locations of the (polymers) component. ▪	 FIGURE 2: Pattern resulted from interference of two beams on stainless steel[2]

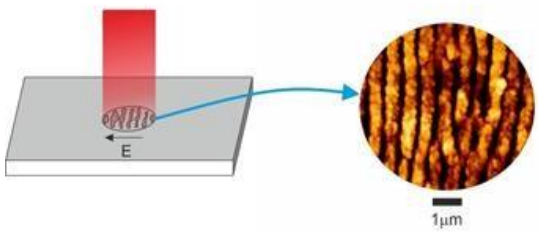
Process speed is a key feature of a laser functionalisation process to achieve its up-scalability into mass production lines. Besides, it has been widely demonstrated that for certain applications (including tuning the optical properties[2] of materials or biological applications[3] among others) the size of the textures induced in the surface of the material needs to be in the order of a few hundreds of nanometres. Therefore, **neither DLA nor DLIP approaches are adequate for the definition of submicron structures in metals.**

In this context, surface functionalization with ultrashort pulses via Laser Induced Surface Periodic Structures (LIPSS) emerges as an optimum candidate for metal industrial applications. LIPSS are periodic structures naturally created by the interaction of ultrashort pulses with a surface, at the vicinity of the material ablation threshold. High resolution features (in the range of 150-800 nm) can be defined in precise locations of the component.



TODAY ACHIEVABLE TECHNOLOGY DEPENDENT SURFACE FABRICATION SPEED AS A FUNCTION OF STRUCTURE SIZE

In the last decade, LIPSS have attracted increased research attention because of their applications in surface hydrophobic/hydrophilic properties, friction reduction, control of the surface reflection, control of the cell growth direction or Surface-Enhanced Raman Spectroscopy. LIPSS have been induced in a wide variety of materials (metals, semiconductors, polymers and ceramics) by pulsed lasers [4,5].



Laser systems in the range of ultra-short pulses (≤ 10^{-12} sec.) offers a set of advantages for the functionalisation of metallic surfaces via LIPSS. For instance, this type of laser provide flexibility to modify a limited area of the manufactured piece or to apply different functions in different areas in the same piece and are also suitable to work on 3D complex pieces or to process different materials. Furthermore, laser systems are especially suited for mass customisation due to its versatility. However, ultra-short pulse laser systems are currently only used in small series manufacturing processes of high value components, not being feasible yet for the mass manufacturing processes due to:

- As for **laser technology** (equipment and process):

[2] Rahul Dewan, Dietmar Knipp, Light trapping in thin-film silicon solar cells with integrated diffraction grating, Journal of Applied Physics 2009 106:7
[3] E. Stratakis, et al., Biomimetic micro⁄nanostructured functional surfaces for microfluidic and tissue engineering applications, Biomicrofluidics. 2011, 5(1): 013411
[4] A. Pan, et al., Formation of laser-induced periodic surface structures on niobium by femtosecond laser irradiation, J. Appl. Phys. 115 (2014) 173101..
[5] Jorn Bonse et al., Laser-Induced Periodic Surface Structures— A Scientific Evergreen, IEEE Journal of Selected Topics In Quantum Electronics, Vol. 23, No. 3, May/June 2017

  - Total Cost of Ownership (TCO) is high due to the investment costs but also due to the maintenance cost of optics and consumables (diodes).
  - For 3D large size pieces, time processing is long [6] (up to 30 min/cm^2). This is a strong constraint for its use in mass production.
- **In-process inspection** is not available for those functional properties arising from sub-micron structures (such as coating thickness and depth profile), thus making impossible to evaluate the quality of the textured surfaces and therefore the functional properties derived from these structures.
- There is a need of **simulation process** tools to ease the definition of laser parameters. Currently, there is not a methodology to set-up the technical working conditions, being necessary the realization of different experiments to determine the relation between parameters and the structure required.

LASER4SURF will **overcome the drawbacks that avoid the use of LIPSS on metallic surfaces at mass production level**. This will be achieved by developing a Laser texturing based system for the surface functionalization of metals using ultra-short laser sources to create the following product functionalities: wettability/lubrication/anticondensation; enhanced/reduced adhesion; osteointegration/antibacterial; soft-touch; tuning optical properties (antireflection (black metals) and aesthetics) properties.

LASER4SURF consortium involves 2 research organisations partners and 2 industrial partners, covering all the fields of expertise required for the fulfilment of the objectives of the project: laser technology, in-line inspection process and simulation process. Besides, the project includes a communication partner and 3 end users including, global leading companies in machine tool industry, health care sector and battery sector, which will facilitate the validation of the project results and the future entrance into the market.

1.1 OBJECTIVES

LASER4SURF **main objective** is to remove the technical and economic barriers that prevent the use of laser technologies for obtaining functionalised metallic surfaces with textures ~1µm or less on mass production. This objective is mainly achieved by the development of an integrated Laser Texturing Prototype that comprises laser texturing device, in-line monitoring system and simulation tools.

The **specific objectives** pursued by the project are:

- To develop a laser-based system that allows the industrial functionalization of parts of up to 300 cm x 100 cm x 20 cm.
- To develop a laser-based system that allows to obtain the following functionalities on metallic surfaces: wettability/lubrication/anti-condensation; enhanced/reduced adhesion; osteointegration/antibacterial; softtouch; tuning optical properties (antireflection (black metals) and aesthetics) at industrial level. ▪ To reduce a 95% the time for the functionalization process of surfaces down to 0.1 min/cm^2 ▪ To demonstrate an increase in the product performance of 3 uses cases as follows:
  - Batteries:
    - Improve a 30% the life cycle of battery electrodes, compared to electrodes casted on conventional current collectors.
    - Increase the battery C-rate capability by 20% when cycling at 1C and by 60% when cycling at 2C.
  - Encoders:
    - To **increase twofold** the tolerances of distances between the grating and the optical head
    - To **eliminate the background light** of the gratings by incorporating polarization management
  - Implants: Improve the implants adhesion by **25% (7.5 N/cm torque)** after one month and by 80% (45 N/cm) after three months.

These objectives will be achieved thanks to the development of the following **operational objectives**:

- To develop a Laser Texturing Prototype based on the definition of Laser Induced Periodic Surface Structures (LIPSS) using ultra-short laser sources (500 fs). This prototype will be developed to reach texturing surfaces with a lateral width of ~1µm or less with precise control of orientation of the structures
- To develop an In-line processing monitoring unit based on Optical Diffractometry, able to ensure the quality requirements in the sub-micron range.

[6] G. Mincuzzi, L. Gemini, M. Faucon, R. Kling, Extending ultra-short pulse laser texturing over large area, Applied Surface Science 386 (2016) 65–71

- To define a Modelling Tool that links the process parameters with the surface modification and the targeted functionality.
- To integrate the developed Tools (laser texture prototype + in-line processing monitoring system+ modelling tool) into a Prototype System to obtain a functional system prototype.
- To validate the tools and system developed in 3 different use cases: advanced batteries, linear encoders and medical components, including the analysis of the productivity and economic and environmental feasibility of the processes (TRL 6).

1.2 RELATION TO THE WORK PROGRAMME

LASER4SURF is fully aligned with the main aspects and characteristics of the topic FOF-06-2017 "**New product functionalities through advanced surface manufacturing processes for mass production**". The relevance of LASER4SURF to the specific topic is explained in the following table:

FOF-06-2017	LASER4SURF proposal
SPECIFIC CHALLENGE	
As a response to increasing competition in global markets, many industrial sectors (e.g. automotive, aerospace, tooling or packaging) aim at improving their product performances through surface functionalisation.	To improve the product performances, LASER4SURF project will work on the development of a Laser Texturing Prototype for the functionalisation of metal surfaces for mass production.
As the products are increasingly complex in terms of scale (from nano to macro) and shape, processes need to deliver efficiently, ensuring an uncompromised quality together with high versatility and controlled costs. One way to reach this goal is to differentiate between a product body and its surface, where specific properties can be tailored.	The LASER4SURF project will develop surface functionalisation Prototype, based on LIPSS. The use of laser technology will improve the flexibility and versatility of the solution, enabling the modification of a limited area of the manufactured piece, easing the application of different functionalities in different areas of the same piece and making it suitable to work 3D complex pieces
Furthermore, the required functionalities may be achieved with little or no addition of new raw material. For example, modifications in the surface geometry or even microstructure induced by texturing processes enable to improve the performance of those products by providing them with dedicated functionalities such as tailored friction, antibacterial properties, aesthetic issues or self-cleaning capabilities, among others.	**LIPSS** are periodic **structures naturally created by** the **interaction of ultrashort pulses with a surface**, **no addition of material is required**. The system will allow to obtain multiple LIPSS structures with different sizes and orientations, that can be combined to get different functionalities. The project will focus on the following functionalities: Wettability / Lubrication / Anticondensation *(corrosion resistance)*; Osteointegration / Antibacterial; Soft-touch; Optical properties (diffraction, reflection, aesthetics); Enhanced/reduced adhesion *(surface friction increase or decrease)*
In this context, substantial research is needed for exploring innovative approaches aimed at producing high added-value functional surfaces by a superficial modification of the substrate	The LASER4SURF project will work on the development of an optical module to reach texturing surfaces of ~1µm or less with precise control of orientation of the structures on mass production.
Special attention should be paid to the cost efficiency of the novel surface manufacturing processes and to the development of technologies that are adaptable and up-scalable to real scale conditions and to their implementation into mass production conditions. Finally, environmental aspects of the processes should also be addressed.	The LASER4SURF project will focus on the development of three different technologies, to solve the three main limitations of Laser texturing to reach mass production: (i) Process time; (ii) Lack of inline inspection. (iii) Lack of a methodology for the parameters selection.
SCOPE	

The proposal should address surface-modifying methods which do not alter the chemical composition of the surface or add an extra layer of a different material, for example: micromachining, texturing, photon-based technologies, laser, mechanical treatments, etc.	The LASER4SURF project will develop a LASER Texturing Prototype, for surface functionalisation by LIPSS. As mentioned before with this technology, no addition of material is required.
These methods should be used to create new manufacturing processes that can be applied on mass production lines.	The main objective of the LASER4SURF project is to develop a laser texture system that will allow the implementation of LIPSS process in production lines.
Due to the need for cost-effective technologies, these processes should be easy to integrate within the existing manufacturing plants and costeffectiveness should be demonstrated.	The LASER4SURF solution is conceived as an independent system and therefore, bringing flexibility to integrate it in existing plants. The cost effectiveness of the Laser Texturing Prototype developed in the project, will be demonstrated for three different industrial use cases (advanced batteries, linear encoders and dental implants).
The research activities should be multi-disciplinary and address all of the following issues:	LASER4SURF research activities cover all the issues as detailed below:
▪ Development of cost-efficient, up-scalable and adaptable surface processing techniques that introduce micro- or nano-scale modifications at the surface level of the part providing it with specific properties or capabilities.	The project will develop a multi beam optical module, to reduce processing time, enabling its use in a wide range of sectors and easing its up-scalability. Moreover, LIPSS technique will be developed to reach texturing surfaces with a lateral width of ~1µm or less with precise control of orientation of the structures
▪ Design and implementation of specific methods and systems that enable highly efficient up-scaling of the developed	The Laser Texturing Prototype will incorporate a modelling tool to facilitate the selection of the process parameters

processing techniques from laboratory scale to real scale, with a specific objective to apply the processes for mass production.	
▪ Implementation of modelling tools to support selection of the processing parameters that lead to the targeted surface modifications.	LASER4SURF project will develop a modelling tool (WP4) that will enable industrial end-users to select the process parameters that give rise to a desired functionality, without a need to deeply understand the LIPSS generation process
▪ Solutions which are economically viable, environmentally friendly and easy to transfer to other fields than the demonstrated fields of application.	The LASER4SURF project will demonstrate the cost effectiveness of the Prototype developed for three different industrial products. To ease the replication of the project results into other fields, will work under WP7 to define new strategies for an effective replication of the process developed, not only in the specific sectors cover by the uses cases, but also in other sectors.
▪ In-process inspection and monitoring possibilities to ensure that the final results remain within the quality requirements	Under WP3 the project will develop an inline monitoring process that will be integrated in the global solution.

1.3 CONCEPT AND METHODOLOGY

1.3.1 OVERALL CONCEPT UNDERPINNING THE PROJECT

LASER4SURF project will address the development of an innovative Laser Texturing Prototype for the surface functionalization of metals using ultra-short laser sources. This Prototype will be based on the development of a cost effective and up-scalable technology, to enable its incorporation on mass production lines.

The laser based system for the functionalization of metals to be achieved in LASER4SURF is conceived:

- To generate LIPSS of different pattern sizes with multiple orientations. These structures and orientations could be combined to obtain new or tailored functionalities.
- To reduce the time processing in 95%.
- To allow in-line processing inspection of submicron structures. ▪ To ease the definition of the process parameters.

The development of this system will be focused on three modules that will be integrated in one single prototype:

- **Laser technology/equipment**. To generate LIPSS of different pattern sizes with multiple orientations, LASER4SURF will develop an easy to integrate and control, compact and low cost optical module. This optical module will be the responsible for the control of the parameters that will determine the LIPSS formation: optical power, beam profile, polarisation orientation, and even the operation wavelength of the micromachining system.
- **In-line inspection tool**. A whole optical measurement system based on diffractometry technology will be integrated with the laser equipment including hardware and software in order to be able to monitor all different nano properties from the functionalised surfaces.
- **Simulation tool** that will show automatically the process parameters required to obtain the desired functionality for a specific material. The simulation tool will automatically transfer the optimum laser parameters to the LIPSS equipment in order to configure it for the desired pattern. These parameters include laser energy, laser polarization, scanning speed, line overlap and beam shape. It will be integrated inside the LASEA proprietary software and will be transferred to the Optics module, in order to configure it for the desired process. The simulation tool resulting from the project will enable the end-user to select a specific functionality from a previously defined database. This database could be updated and enlarged after project end for functionalities and/or materials different from the ones selected for the project duration.

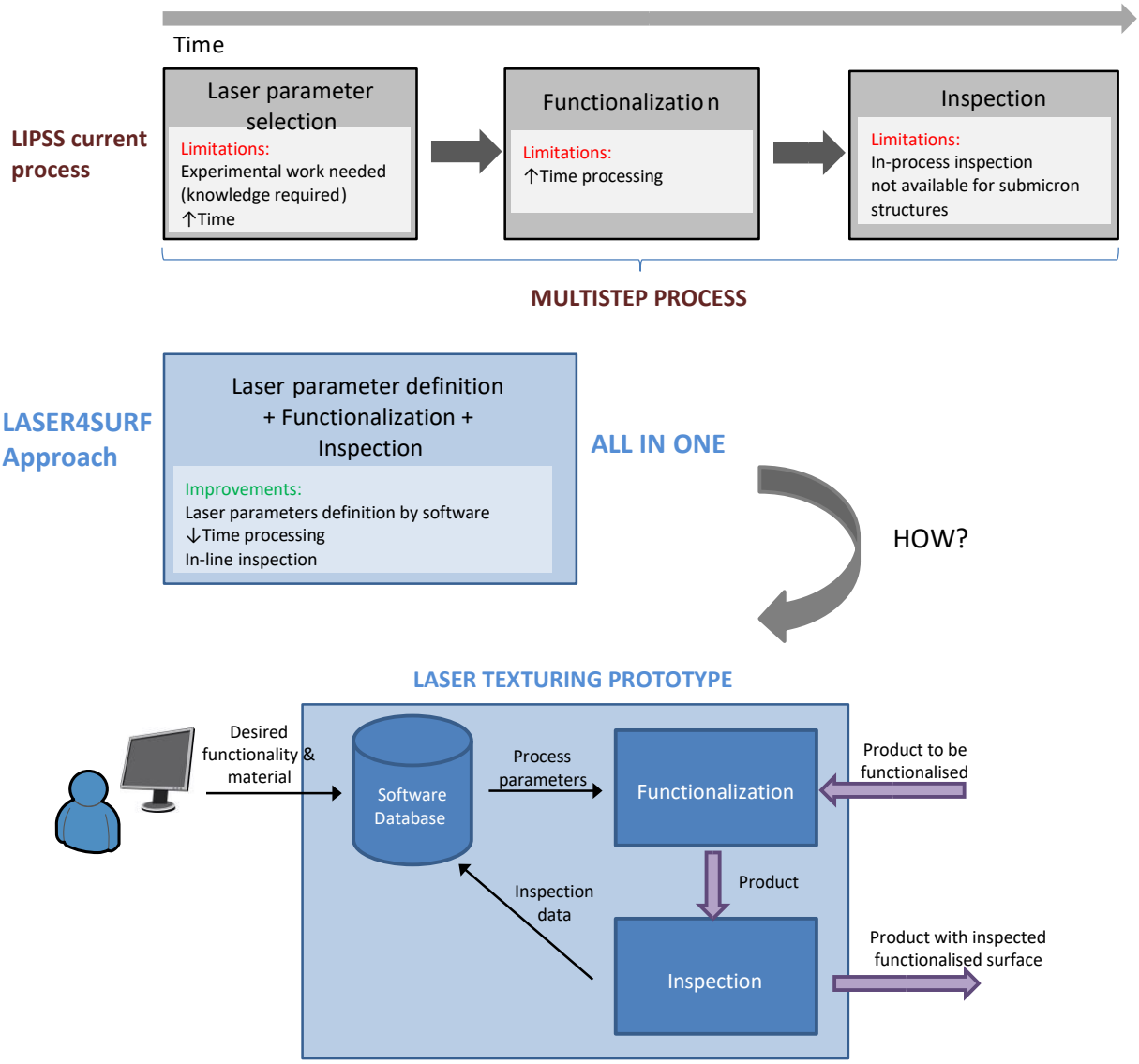


FIGURE 3: LASER4SURF CONCEPT

MACHINE

LASER4SURF new Laser Texturing Prototype will merge all the surface functionalisation independent process: **definition of process parameters + functionalisation + inline inspection, in one single process.**

The LASER4SURF Laser Texturing Prototype will be a cost-effective solution for the surface functionalisation of metals on mass production. To assure the up-scalability of the Prototype to any industrial process, the project will cover a representative panel of metal materials (Aluminium, Copper, Titanium and Stainless steel) and functionalisation. The selection of these materials is made to facilitate its replication in other fields than the ones demonstrated in the LASER4SURF project (biomedical, advanced batteries and machine tool). The future implementation of the project results will be assured as these materials:

- Are the most commonly used in industrial applications, ranging from installations, heat exchangers, tools, and many others.
- Cover a wide range of thermal conductivity values:
  The general approach for process time reduction relies on increasing the power and/or the repetition rate of the laser source. However, for femtosecond laser processing of materials, an incubation effect has been reported for high repetition rates. This effect, that results in a modification of the ablation threshold as the number of pulses increases and may be translated into an increased processing speed, highly depends on the thermal properties of materials[7].

[7] Wagner de Rossi et al., Effects of ultrashort laser ablation in copper and stainless steel, Lasers in Manufacturing Conference 2015

Therefore, optimizing the repetition rate and the accumulated pulses for materials with very different thermal properties will enable the LASER4SURF results to be easily extrapolated to most of the metals alloys currently used in manufacturing industry, assuring its future implementation in any industrial sector.

The four materials selected possess very different thermal properties (see Figure 4), covering a wide range of thermal conductivity values (from highest values such as 386 /m K of Copper to the lowest: 19-23 of Titanium). Despite the difference in the thermal conductivity among materials, the necessary laser parameters to achieve a required patter size in metallic materials differs slightly from one metal to another (laser fluence from 0.1J/cm^2 for Titatium to 0.6 J/cm^2 for Cu for a single pulse[8]) in contrast to other materials, where fluences as high as 8J/cm^2 are needed[10]. The range of required fluences are easily obtained from a single commercially available femtosecond source, that usually deliver high powers, thus enabling the extrapolation of LASER4SURF results to other metallic materials.

Metal	Thermal Conductivity- *k* -	
	(Btu/(hr °F ft))	*(W/m K)*
Aluminum, pure	118	204
Aluminum Bronze	44	76
Carbon Steel, max 0.5% C	31	54
Carbon Steel, max 1.5% C	21	36
Chromium	52	90
Copper, pure	223	386
Nickel	52	90
Niobium (Columbium)	30	52
Platinum	42	73
Silicon	48.3	84
Silver, pure	235	407
Stainless Steel	7-26	12 – 45
Tin	36 - 39	62 – 68
Titanium	11 - 13	19 – 23

FIGURE 4: THERMAL CONDUCTIVITY OF DIFFERENT METALS

The solution will be developed to obtain multiple LIPSS structures with different sizes and orientations, that can be combined to get the different functionalities. To demonstrate the viability of the solution suggested by LASER4SURF, the project will focus on the following functionalities for each of the selected materials (see table X): Wettability / Lubrication / Anti-condensation; Enhanced/reduced adhesion; Osteointegration / Antibacterial; Softtouch; Tuning optical properties, including Antireflection (black metals) and Aesthetics.

Functionalities	Aluminium	Copper	Stainless steel	Titanium
Wettability / Lubrication / Anticondensation *(corrosion resistance)*	X	X	X	
Osteointegration / Antibacterial			X	X
Soft-touch	X	X	X	X
Optical properties (diffraction, reflection, aesthetics)			X	
Enhanced/reduced adhesion *(surface friction increase or decrease)*	X	X	X	X

These functionalities can be applied to many different sectors, such as: Automotive, Electronics, Electrical engineering, Environmental technology, Lighting, Medical technology and biotechnology, Sensor technology and Tool making and mechanical engineering. Three different use cases representing three different industrial sectors, covering different materials, product sizes and functionalisation required will be considered in LASER4SURF: energy storage, biomedical and machine tool. The uses cases and its desired functionalisation are the following: (i)

[8] K. M. Tanvir Ahmmed, C. Grambow and A.M Kietzig, Fabrication of Micro/Nano Structures on Metals by Femtosecond Laser Micromachining, Micromachines 2014, 5, 1219-1253;
[10] G. Dumitru, V. Romano, H.P. Weber, M. Sentis, and W. Marine, "Femtosecond ablation of ultrahard materials", Appl. Phys. A, 2001, 74, 6, pp. 729-739.

Encoders: Tuning optical properties, (ii) Implants: Osteointegration / Antibacterial and (iii) Batteries: Enhanced/reduced adhesion (surface friction increase or decrease).

1.3.2 POSITIONING OF THE PROJECT

The activities within the present proposal will span between TRL 4 to 6. The project will start as a TRL 4 (pilot study will be validated at lab-scale showing that a wider and integrated optimisation of the value chain as a whole, could have the potential to reduce cost) trough TRL 5(the value chain components can be used individually but not in an integrated and optimized way) leading to TRL 6 (Representative Laser Texturing Prototype system is tested in a relevant environment, simulating the operation conditions of the three use cases.).

1.3.3 NATIONAL OR INTERNATIONAL RESEARCH AND INNOVATION ACTIVITIES

Several previous national and international R&D activities that will be linked to LASER4SURF have been carried out/are being carried out.

PREVIOUS PROJECT	Achievements/outputs relevant to LASER4SURF	Innovations of LASER4SURF
FRUITFUL "Flexible and Robust processing Units based on Innovative amplification Techniques of Fibre Ultrashort Lasers" EUROSTARS, E! 7701. Partner involved: MULTITEL	Development of ultrashort (500 fs) high energy and compact laser source for material processing.	Potential seed laser source that could be used in LASER4SURF for process developments. Combined with an amplification stage, such source could be a candidate for processing any of the metallic surfaces targeted in the project.
LASHARE "Laser equipment shaping ASsessment for High impAct module. innovation in the manufactuRing (FP7-2013-NMP-ICT-FOF, n°609046) www.lashare.eu Partner involved: MULTITEL, LASEA	Development of a high-energy picoseconds laser system for solar cells processing. Fully prototyped version integrated in LASEA premises.	Integration of a beam with online control European industry"
PMJOIN: Development of a direct laser joining of hybrid plastic-metal components for industrial applications. EU-FP7. http://www.pmjoin.eu	Development of laser textures for the hybrid joining of metals with polymers.	Improvement in the generation of laser surface textures.
ALPINE "Advanced Lasers for Photovoltaic INdustrial processing Enhancement" (FP7-NMP-2008LARGE-2, n° 229231) www.projectalpine.eu Partner involved: MULTITEL	Various materials (Molybdenium, CIGS, CdTe, ZnO) process developments for thin film and flexible solar cells applications. This project was mainly focused on nanosecond and sub-nanosecond laser sources. Top-hat beam shaping methods were used for clean process definition.	LASER4SURF includes the use of ultrashort lasers and in particular LIPSS process.
ADOPT "ADaptive laser micro machining of multi-layer surfaces with inline OPTical metrology" CORNET, convention n°1510569 Partner involved: MULTITEL	Development of an Optical Coherent Tomography (OCT) online control system. This system is dedicated to the control of micron and tens of microns features.	The features in LASER4SURF will be micro and nanoscale.

		New optical properties like depth profile will be monitored under LASER4SURF project.
PREVIEW "PREdictiVe system to recommend Injection mold sEtup in Wireless sensor networks. (nº 636892). Partner involved: VISUM.	New predictive models and data acquisition systems are obtained for monitoring processes.	LASER4SURF will make a step further improving the monitor systems with a diffractometry technology able to monitor Nano textures.
"Cloud-enabled IR spectroscopy integration architectures monitoring system solutions for food and pharmaceutical industries"	obtained from the and real-time chemical monitoring 2º Phase SME Instrument Partner involved: VISUM.	The software and hardware platform for the rapid and customisation of affordable, VISUM contribute to the success of the LASER4SURF project.

PREVIOUS PROJECT	Achievements/outputs relevant to LASER4SURF	Innovations of LASER4SURF
LASERHUESO: "Laser fabrication of Surface micro/nanopatterns for the optimization of osteocells regeneration potential", CYCIT 2014 (Spanish Science Foundation) Partner involved: CEIT	The global objective is to produce tailored patterns from biocompatible materials (Si, Ti, and biomedical polymers) using ion- and/or laserbased techniques for the development of therapeutic	approaches to bone restoration Laser patterned surfaces will not be limited to planar substrates and/or biocompatible materials.
ECOGRAB: Design of a prototype for industrial engraving of diffraction/phase gratings for encoders applications using lasers Partner involved: CEIT, FAGOR	The objective of the project is to stablish the limits of laser patterning of gratings as well as the design of an optimum grating configuration (size, period, depth…)	LASER4SURF will provide In-line monitoring capabilities for the prototype as well as the support for parameter selection. Additionally, the final prototype will be implemented and tested for this application
SIRBATT "Stable Interfaces for Influence of battery interfaces in the Up to now, the cycle life and Rechargeable Batteries" (FP7- cycle life and safety of Li-ion batteries safety of Li-ion batteries have ENERGY-2013-7-3-3, n° 608502) been mainly correlated with the Partner involved: CICe electrode material. LASER4SURF will unveil the influence of the current collector surface on the battery safety and cycle life.		

1.3.4 OVERALL METHODOLOGY

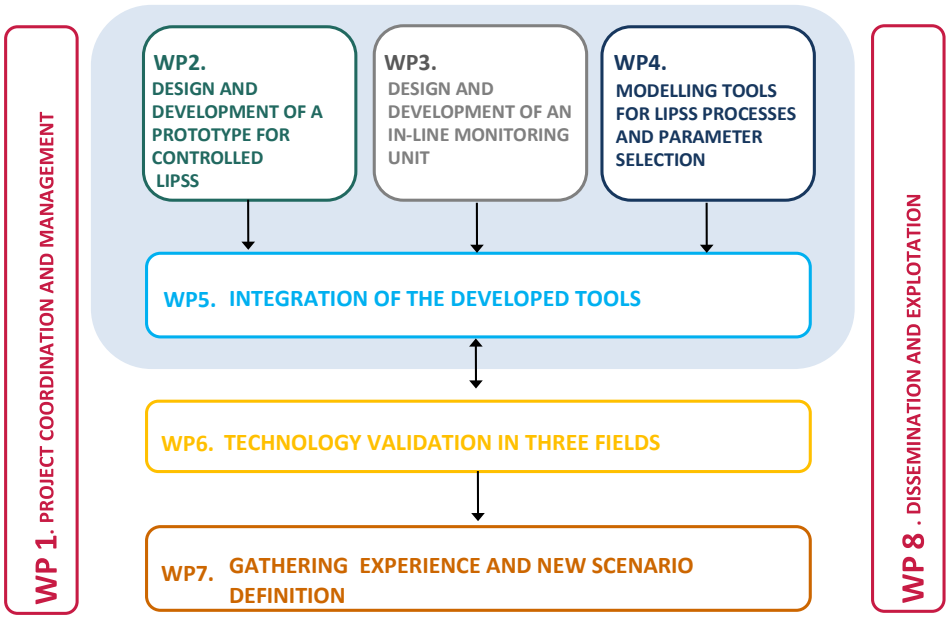
To achieve LASER4SURF goal and the specific objectives, the project will be developed in 3 different phases structured into 8 work packages. Besides the cross-cutting Work Packages, WP1 Project Management and WP8 focused on Dissemination and Exploitation:

The **first phase** (WP2-5) aims to develop each one of the different modules that will be included in the LASER4SURF system and to integrate the developed modules in one prototype:

- *WP2. Design and development of a prototype for controlled LIPSS*. Within this WP, the optical module that will permit to control the optical power, beam profile, polarisation orientation and the operational wavelength of the micromachining system will be developed
- *WP3. Design and development of an in-line monitoring unit.* An optical measurement system, based on diffractometry technology, will be developed. This module will be able to monitor all different nano properties from functionalised surfaces.
- *WP4. Modelling tools for LIPSS Processes and Parameter Selection.* WP 4 is devoted to develop a software platform that will enable industrial end-users to select the process parameters that give rise to a desired functionality, without a need to deeply understand the LIPSS generation process. The development of this tool is divided into four steps: i) generation of a process database, ii) statistical Analysis; iii) design and development of the SW platform and, iv) pre-validation of the SW tool.
- *WP5 Integration of the developed tools.* The development of WP2, WP3 and WP4 will lead to **WP5**. In this WP, the consortium will work on the interconnection between the different modules developed in previous work packages. The different modules developed in WP2, WP3 and WP4 will be integrated on an already existing setup. Different validation tests will be carried out in order to check and compare the functionality of every module.

Second phase: Technical validation and demonstration of the developed innovative solution will be done at operational environment (*WP6. Technology validation in three fields*). The solution will be validated using products of three different use cases, focussed on different sectors, with different requirements in terms of surface functionalization (See below for more detailed information about each one of the use cases). All the results obtained during the validation phase will be analysed and incorporated in the system developed in previous WPs in order to improve it.

Third phase: *WP7 Gathering experience and new scenario definition.* With the aim of maximising the future impact of the project results, this WP aims to compare the results obtained with the expected results in order to define new strategies for an effective replication of the process developed in other areas of application within the sectors covered by the use-cases as well as in other sectors.

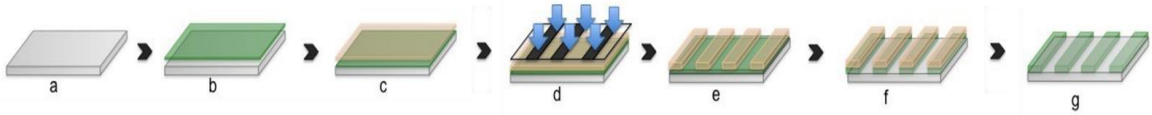


1.3.4.1 USE CASE DESCRIPTION

The cost-effectiveness and up-scalability of the new Laser Texturing Prototype will be demonstrated and validated upon three simulation operational environments, representing three different industrial sectors, covering different materials, product sizes and functionalisation required.

MACHINE TOOL

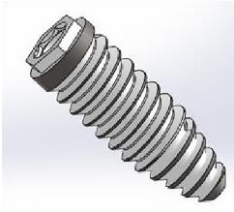
Linear encoder - Use case leaded by: FAGOR

General description of the product
A linear encoder is a sensor, transducer or readhead paired with a scale that encodes position. Optical encoders are the most accurate of the standard styles of encoders, and the most commonly used in industrial automation applications. An optical encoder includes two identical linear gratings: one fixed (ruler or scale) and the other one mobile (scanning head or reticle), which are illuminated by a LED. The measurement process of an optical linear encoder entails photoelectric sensing of the scale and a scanning reticle that move relative to each other.
Current surface functionalisation of the product
Lithographic process and wet etching for the definition of the grating in the Cr layer
General description of the process currently used to modify surface characteristics:
Photoresist assisted lithographic process and wet etching such as the one presented in the figure below:  FIGURE 5: LITHOGRAPHIC PROCESSS A) SUBSTRATE B) CR THIN-FILM DEPOSITION C) PHOTORESIST DEPOSITION D) MASK LITHOGRAPHY E) PHOTORESIST DEVELOPMENT F) CHEMICAL WET ETCHING G) PHOTORESIST CLEANING.
Problems/Limitations of the current product/process
▪ Environmental problems (use of chemicals and toxicity of Cr). ▪ Accuracy problems, due to dirt accumulation. ▪ Low resolution and accuracy problems due to gratings subjections to Talbot effects.
New or improved functionalisation proposed in LASER4SURF project
▪ Wettability / Lubrication / Anti-condensation. Avoid the accumulation of dirt. ▪ Tuning optical properties. Change the polarization of the reflected light
Problems/Limitations to solve
▪ Makes the optical encoder more robust to miss alignments. The distance between the grating and the optical head with state-of-the-art technology must be kept constant within a tolerance: ▫ For transmittance encoder systems the distance is 200um and the tolerance ±10um ▫ For reflective encoder systems the distance is 1.5mm and the tolerance ±0.15mm. ▪ We want these tolerances to increase twofold with this technology ▫ The tolerances of the mechanical assembly for the final product can be increased, reducing the aggregated cost of the complete system. ▪ The optical contrast will increase by elimination the background light, Optical contrast is the relation between the minimum and maximum optical signal. This signal is always added to a background light that produces a DC offset of the electrical signal. Polarization management will eliminate this signal and simplify and make more robust the electrical conditioning of the optical signal.

BIOMEDICAL

The LASER4SURF project counts with the biomedical implant developer RESCOLL and producer RESCOLL Manufacturing (Third Party of RECOLL), who will work on the Prototype validation, using dental implants.

Dental Screw - Use case leaded by: RESCOLL
General description of the product

Dental screws for implantation prior to attachment of artificial replacement tooth implants. The screws are manufactured of titanium alloy (Ti6Al4V) using traditional methods (milling, cutting, polishing).	
Current surface functionalisation of the product	
None. Only cleaning and polishing processes are applied on the surface prior to quality control.	
General description of the process currently used to modify surface characteristics:	
Current implant processing does not use any particular method of surface postprocessing. Only cleaning and polishing processes are applied on the surface prior to quality control.	
Problems/Limitations of the current process	
No microtexturing or surface postprocessing is currently used.	
New or improved functionalisation proposed in LASER4SURF project	
Cellular adhesion and osteointegration.	
Problems/Limitations to solve	
▪ Very precise dimensional tolerances (~1-2 µm) ▪ Must not modify the nature of the metal ▪ Complex geometry to treat – screw threads must remain undamaged ▪ Small size	

The improvement on Cell adhesion and osteointegration achieved by LIPSS will be proven for dental implants during the project. Besides during WP7 the replication on other implants (like hip replacements or spinal screws) that have regulatory restrictions far more severe, and will need more tests, and longer testing times, what means a longer time to market implementation.

BATTERIES

Batteries - Use case leaded by: Cegasa (Third Party of CICe).	
General description of the product	
The products to be functionalised on the project are current collector for coin cells 2032.	
Current surface functionalisation of the product	
Coating of the metallic surface with a carbon layer.	
General description of the process currently used to modify surface characteristics:	
The coating of the metallic surface with a carbon layer, it is made by means of chemical vapour deposition (CVD) techniques. The main point of this procedure is the generation of a new interface (C-metal) that will enable high power density batteries.	
Problems/Limitations of the current process	
▪ C-coat degradation. ▪ Increase of the internal resistance of the cell. ▪ Poor adhesion of the electrode.	
New or improved functionalisation proposed in LASER4SURF project	
Enhanced adhesion and roughening of the current collector (improving the adhesion of the coating and therefore the number of cycles). Increase electrode mass loading per cm2 and decrease of diffusion paths hence improving current density performance.	
Problems/Limitations to solve	

- Planarity of the current collector surface will have to be very good in order to match the laser interference pattern
- Presence of impurities in the current collector surface previous to the LIPSS processing might lead to formation of undesired carbonaceous species during processing.
- Control of undesired electrochemical side reactions that will appear due to the increase of effective surface area in the current collector.

The improvement of the current collector performances (increase the mass loading and decrease of diffusion paths hence improving current density) will be validated under industrial requirements of coin cells. Besides, under WP7 the project will study the implementation of the process for larger batteries, focusing in current collectors for pouch cells for electric vehicle, a growing market with the capability to increase the current production 20 times by 2040 (reaching 48000 km^2/year of collectors surface).

1.3.5 SEX AND/OR GENDER ANALYSIS

The project's consortium will ensure gender balance whenever possible and to promote participation of women in the different tasks of the project. Some of the specific actions that will be taken to ensure gender balance are:

- General management of the project will guarantee gender equality, i.e. equal consideration to the life patterns, needs and interests of both women and men;
- LASER4SURF project will make every possible effort to gender mainstreaming in order to give European women an equal chance of shaping, participating and monitoring EU supported scientific projects and programmes;
- Flexible working-time arrangements will be promoted among participating organisations for men and women to reconcile work and private life, not interfering in terms of income and career perspective.

In addition, it is important to highlight that gender balance is also an individual commitment for each one of the partners as it is included in their internal policies and in line with national regulations.

1.4 AMBITION

1.4.1 ADVANCE BEYOND THE STATE-OF-THE-ART

In the following paragraphs, the current state of the art, the limits and barriers as well as the advance beyond the state of the art of the methods and systems proposed in LASER4SURF are explained. A summary of the main advances is included below:

Solution	State of the art	Beyond the State of the art
Parameter selection.	There is not a methodology for the selection of laser parameters in order to obtain LIPSS structures.	Development of a tool to predict and optimize the selection of the process parameters
LIPSS Combination of a high repetition rate laser Multiple beams processing, taking advantage of **process.** with a galvanometric scanner and Gaussian the low energy per pulse required for this process. beam shape The LASER4SRUF combination would consist on using a high repetition rate laser source delivering high average power (range of 100 W typically) and		split over several beams (up to 10) delivering a few microjoules per beam.
Inline inspection.	Off line implemented techniques (SEM and AFM) require sample preparation Current approaches calculate the spatial period by estimating line thickness and spacing line by line, and generating a mean value.	Optical techniques for inline process monitoring, as they are non-contact, non-destructive and can be applied to samples as they are, without pretreatment.

Laser technology/LIPSS

State of the art:

Laser-induced periodic surface structures (LIPSS or ripples) is a universal phenomenon observed on solids upon irradiation using linearly polarized laser pulses[9]. LIPPS have been identified as a universal phenomenon that can be observed on all classes of materials when irradiated near their ablation threshold[10].

LIPSS are periodic structures naturally created by the interaction of ultrashort pulses with a surface, at the vicinity of the material ablation threshold. The orientation of the structures depends on the laser polarization and the period is directly related with the wavelength of the laser source.

Two distinct types of LIPSS have been identified: Low Spatial Frequency LIPSS (LSFL) and High Spatial Frequency LIPSS (HSFL) that appear under different laser fluence conditions and that are respectively perpendicular and parallel to the polarization orientation. LSFL have a spatial period close to or slightly smaller than the irradiation laser wavelength. HSFL have a spatial period that is much smaller than the laser wavelength.

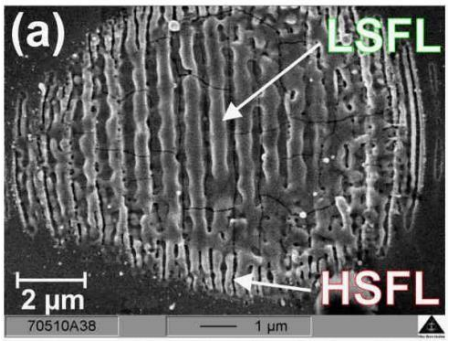


FIGURE 6 EXAMPLE OF LIPSS CREATION WITH LASER CUMULATIVE SHOTS WITH INHOMOGENEOUS FLUENCE REPARTITION ON THE LASER SPOT.

As one can see from Figure 6, beam shaping can have a strong impact on LIPSS formation. As a consequence, polarization, laser fluence and wavelength must be controlled to produce a regular LIPSS structure over a large area for a homogeneous surface behaviour.

Beam shapers

There are different solutions for beam shaping like aperture beam shaping, use of diffractive diffusers, diffractive optical elements, refractive beam shaping or use of spatial light modulators, among other solutions. Each solution has its advantage and drawbacks like losses for aperture-like beam shaping, dynamic flexibility for diffractive optical elements, compactness and optical power handling for spatial light modulators for instance.

Beam shaping solutions	Advantages	Disadvantages
Aperture beam shaping	▪ Simple to use	▪ Important losses
Diffractive optical solutions	▪ Large possibility of beam shapes, single element. ▪ High power handling demonstrated.	▪ More expensive than other passive elements including beam splitting with one
Refractive beam shaping	▪ Efficient solution to produce different beam shapes. ▪ Low loss. ▪ Low cost.	▪ Less compact than a DOE (Diffractive Optical Element) device and not as flexible (multiple beams)
Spatial light modulators	▪ High degree of freedom that can be very useful for laboratory developments. ▪ Not as compact as other solutions.	▪ More expensive than passive solutions. ▪ Optical power damage limit.

For the generation of controlled LIPSS, the steepness and the homogeneity of the beam shape are of great importance to ensure a good compromise between process homogeneity and robustness.

Polarization.

[9] S. Höhma et al., Dynamics of the formation of laser-induced periodic surface structures (LIPSS) upon femtosecond two-color double-pulse irradiation of metals, semiconductors, and dielectrics.

[10] Polarisation-dependent generation of fs-laser induced periodic surface structures. Stephan Gräf, Frank A. Müller Otto-Schott-Institute of Materials Research (OSIM), Friedrich-Schiller-University Jena, Germany

A very important parameter during the LIPSS formation is related to the polarisation state. It is generally accepted that the irradiation with linear polarised laser radiation causes structures aligned perpendicular or parallel to the incident electric field (E-field) vector[13].

Therefore, rotating the laser polarization with optical retarders enables the generation of nanopatterns oriented preferentially to a determined specific direction. This effect can be used to define more complex structures where the direction of the lines is defined independently in every focal point of the surface, giving rise to surface textures like the ones in figure 7.

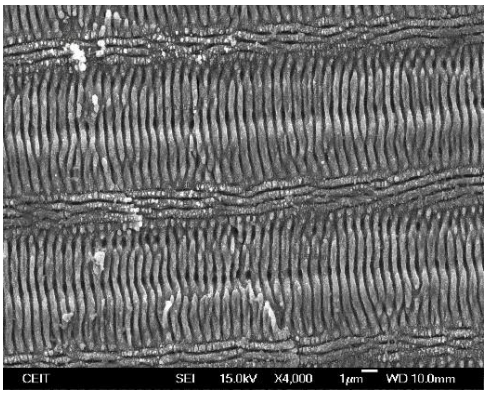


FIGURE 7 SURFACE DEFINITION OF STRUCTURES WITH DIFFERENT ORIENTATION

This approach has been less explored although very promising results have been obtained. For example, for aesthetic purposes, precise control of LIPSS orientation has been exploited for colour marking, as shown in the next figure and cell migration[14].

In the biomedical field, in a very recent work[15], the CEIT research group have demonstrated that nanopatterns oriented in a preferential direction, enable and enhance the control of cell adhesion, migration, and polarization without any chemical treatment in a single step process and human mesenchymal stem cells (hMSCs) tend to attach and preferentially align to the LIPSS nanopatterns oriented in a longitudinal direction.

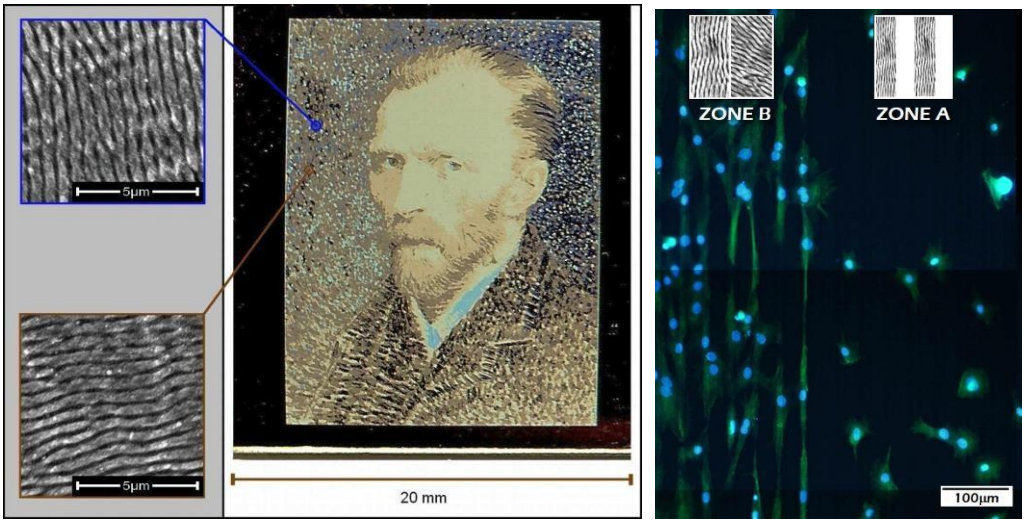


FIGURE 8 A) EXAMPLE OF COLOR EFFECTS OBTAINED BY CONTROLLED NANOSTRUCTURES WITH A FEMTOSECOND LASER AND X6000 IMAGES OF CONTROLLED NANOSTRUCTURES MARKED WITH TWO DIFFERENT ORIENTATIONS[16]

B) THE NUMBER OF CELLS PER UNIT AREA IS TWICE AS LARGER IN ZONE B THAN IN ZONE A BY A FACTOR OVER 2 [17].

A recent work showed the possibility to fully control the orientation of the LIPSS structures by continuously turning the polarisation orientation during the laser beam scanning on the surface:

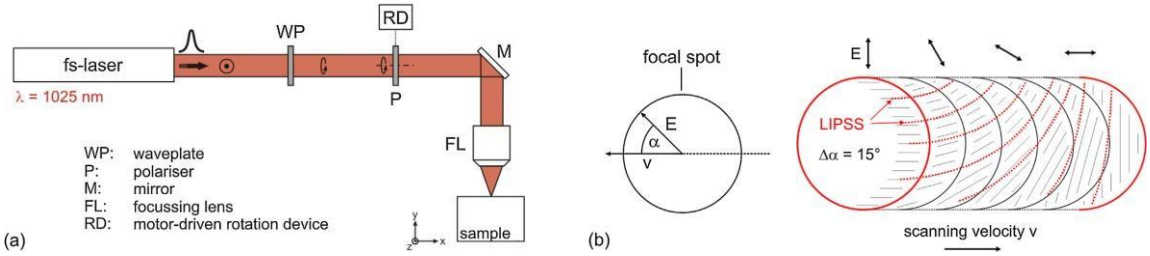


FIGURE 9: (A) Experimental setup and (b) orientation of the e-field vector E relative to the scanning velocity v. Extracted from reference [18].

[13] J. Bonse et al., Femtosecond laser-induced periodic surface structures on steel and titanium alloy for tribological applications, Appl. Phys. a-Mater. 117 (2014)103–110
[14] B. Dusser et al., Controlled nanostructrures formation by ultrafast laser pulses for color marking, OPTICS EXPRESS 18, 3, 2913 (2010)
[15] M. Martínez-Calderon et al., Surface micro- and nano-texturing of stainless steel by femtosecond laser for the control of cell migration, Scientific Reports 6, 36296, 2016
[16] B. Dusser, et al., Controlled nanostructrures formation by ultrafast laser pulses for color marking, OPTICS EXPRESS 18, 3, 2913 (2010)
[17] M. Martínez-Calderon, et al., Surface micro- and nano-texturing of stainless steel by femtosecond laser for the control of cell migration, Scientific Reports 6, 36296, 2016
[18] Stephan Gräf & Frank A. Müller, "Polarisation-dependent generation of fs-laser induced periodic surface structures", Applied Surface Science 331 (2015) 150–155

In this work, the authors presented a simple way to control polarization. Nevertheless, as mentioned by Y. Tang et al.[11], that propose a similar arrangement, some cautions must be taken regarding the quarter-wave plate dispersion that can induce imperfect polarization rotation for broad spectrum ultrashort pulses.

In another type of experiment, the authors used a Spatial Light Modulator (SLM)[12] to control the laser polarization state and induced azimuthally and radially polarized beams:

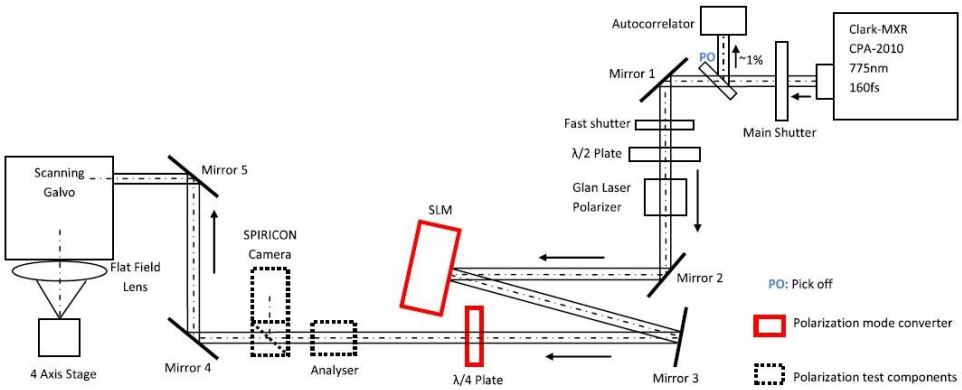


FIGURE 10 EXAMPLE OF EXPERIMENTAL MICROPROCESSING SETUP ALLOWING FOR POLARIZATION MODE CONTROLLING WITH SPATIAL LIGHT MODULATORS.

Apart from the laser beam polarization, there are several process parameters that affect LIPSS formation being irradiation fluence and its distribution (energy per unit area), number of overlapped pulses, irradiation wavelength and repetition rate the most relevant.

Prototype parameter (units)	Effect on Process Parameters	Effect on LIPSS Formation
Laser energy (mJ)	Fluence	Type, depth, period
Laser Polarization	Laser polarization	Orientation
Scanning speed (m/s)	Number of overlapped pulses	Type, depth, period
Repetition rate (Khz)	Number of overlapped pulses	Type, depth, period
Repetition rate (Khz)	Incubation factor and therefore in the ablation threshold of materials	Type, depth, period
Wavelength (nm)	Wavelength (nm)	Period
Laser Beam Shape	Spatial distribution of energy/fluence	Homogeneity of resulting LIPSS

As can be deduced from the table above, some of these determinant parameters mainly depends on the selection of the laser source (energy, repetition rate) and the mechanical parts of the final prototype (scanning speed). However, for LIPSS formation, a polarization control and an appropriate spatial distribution of the laser energy become essential.

Advance beyond the state of the art:

LIPSS is obtained for energy levels very close to materials ablation thresholds. A large number of pulses accumulation is generally needed and a good control of the beam shape and polarisation is needed to ensure an homogeneous patterning. The fastest LIPSS process reported to our knowledge[13] combines a high repetition rate laser with a galvanometric scanner and Gaussian beam shape. The scan direction to the ripples orientation showed an important influence on the texture homogeneity and the global process speed was on the range of cm^2/minute.

LASER4SURF will improve this excellent result by making use of multiple beams processing, taking advantage of the low energy per pulse required for this process, the most promising combination would consist on using a high

[11] Y. Tang et al., "Control of periodic ripples growth on metals by, femtosecond laser ellipticity", Optics Express, 5 November 2012 / Vol. 20, No. 23 / 25826-33.
[12] O J Allegre et al., "Laser microprocessing of steel with radially and azimuthally polarized femtosecond vortex pulses", IOP Publishing, Journal of Optics. 14 (2012) 085601 (9pp),
[13] A. Ruiz et al."High speed inscription of uniform, large-area laser-induced periodic surface structures in Cr films using a high repetition rate fs laser",Optics Letters 39 2014, 2491-4

repetition rate laser source (few MHz rep. rate. typically) delivering high average power (range of 100 W typically) and split over several beams (up to 10) delivering a few microjoules per beam, sufficient for obtaining the LIPSS process.

Such a multi-beam process combined with polarization orientation, fluence and shape control for each beam will be developed in the project LASER4SURF in order to achieve an order of magnitude faster texturation process (on the range of 10 cm²/min) compatible with most of industrial texturing applications and in particular the three use cases of the project.

In-line inspection process

State of the art:

A number of monitoring techniques exist when it comes to characterizing structures in the nano- and micro-metre ranges, and which have been utilised in the development of the LIPSS fabrication technique, including SEM (scanning electron microscopy) and AFM (atomic force microscopy) [14,15].

Fourier transform analysis of the SEM images has been used to quantify the periodicity of the structures. This approach enables a global value of spatial period to be quickly calculated, rather than estimating line thickness and spacings line by line, and generating a mean value, which is a more laborious process.

However, most commonly used techniques (SEM and AFM) at this dimensional scale are in fact implemented off line and require sample preparation. Optical techniques offer the best opportunity for inline process monitoring, as they are non-contact, non-destructive and can be applied to samples as they are, without pre-treatment. According to a recent review paper by IRIS[16], selected combined optical technologies, such as diffractometry, but also ellipsometry and spectral reflectance provide the opportunity for inline measurement of the thicknesses of nanostructured layers.

The main optical techniques currently used are the following: *White light Interferometry:*

Vertically scanning white-light interferometry is a non-contact optical method for surface height measurement on 3-D structures with surface profiles varying between a few micrometers to a few centimeters. Surface features ranging from gradual slopes to vertical steps, with rough or smooth finishes.

An inline spectral reflectance system was developed in the THIME project[25] to measure Organic PV films. White light interferometry has been developed into an inline system by Dr Schwab[17] for measuring the thickness of spacer and cover layers of Blu-Ray discs, but can also be applied to measure the surface topography of reflective materials.

However, while this system measures topography with a high level of accuracy (<100nm), it's lateral resolution is limited to 10s of micrometres. An inline system has been developed by Nanometrics[27] employing spectral reflectometery in single point scanning arrangement, which enables lateral critical dimension measurement with greater accuracy, but still diffraction limited to the spot size of around 1 micrometre.

Scatterometry:

Scatterometry is an optical metrology technique designed for the characterization of test samples from lithography. These samples are diffraction gratings with the width of the lines reproducing the critical dimension. Scatterometry consists in ellipsometric measurements of the light diffracted from the samples in various experimental configurations and the fitting of the data to theoretic predictions. The parameters of the best fit are the most probable values for the parameters of the sample.

Scatterometry involves optical instrumentation, diffraction theories, and data reduction and sensitivity analysis. It is possible to obtain higher resolution in lateral dimension measurements regarding the ones of SEM and AFM techniques, using optical techniques based on scatterometery and diffractometry, which can be applied to periodic structures of the type being developed in Laser4Surf. Scatterometry uses the instrumentation of a spectroscopic

[14] A. Rodriguez, et al, Femtosecond laser-induced periodic surface nanostructuring of sputtered platinum thin films, Applied Surface Science 351 (2015) 135-139
[15] A. Martinez-Calderon, et al., Femtosecond laser fabrication of highly hydrophobic tainless steel surface with hierarchical structures fabricated by combining ordered microstructures and LIPSS, Applied Surface Science 374 (2016) 81-89
[16] E. Bugnicourt et al., Recent Prospects in the Inline Monitoring of Nanocomposites and Nanocoatings by Optical Technologies, Nanomaterials 2016, Review, 6(8), 150 25 THIME, Thinfilm measurements on organic photovoltaic (OPV) layers; FP7Grant Agreement no: 315665; 2012-2014
[17] Dr. schwab GmbH, IQPC – Integrated Quality and Process Control. http://www.schwabinspection.com/index.php?puid=1&pageid=39 27 Nanometrics Impulse, http://www.nanometrics.com/products/IMPULSE.html

ellipsometer, along with complex mathematical modelling[18] and is widely used in the microelectronics industry, for example the Nanometrics Atlas[19], and the Nova T600[20].

However, the complex instrumentation and high power data processing required mean that the technique has not been applied inline. Coherent Fourier Scatterometry has been developed to exploit the smaller spot size that is possible using laser light, but is currently not deployed inline[21]. *Diffractometry:*

Diffractometry using multiple diffraction orders imaged in the far-field is a versatile technique that has potential to provide comprehensive information regarding the 3 dimensional features of textured samples. Furthermore, when combined with different light source, it covers a broad resolution range: UV light can detect structures with period in the few hundreds nm range, whereas cost competitive blue laser can be used for structures with period between 500 and 700 nm. It has been effectively implemented in several versions for characterising the quality of the grooves in DVD and Blu-Ray discs, for example Dr. Schwab's DOMS[22], and Singulus Mastering B.V.[33]

However, there are some barriers to be tackled in order to be able to implement on in-line systems: careful and time-consuming sample preparation are still needed, many samples are vulnerable to radiation damage caused by the incident electrons and the optical system is quite complex to be settled in a typical laser equipment.

Confocal Microscopy:

Confocal microscopy, most frequently confocal laser scanning microscopy (CLSM), is an optical imaging technique for increasing optical resolution and contrast of a micrograph by means of adding a spatial pinhole placed at the confocal plane of the lens to eliminate out-of-focus light. It enables the reconstruction of three-dimensional structures from sets of images obtained at different depths (a process known as optical sectioning) within a thick object. This technique has gained popularity in the scientific and industrial communities and typical applications are in life sciences, semiconductor inspection and materials science. It may be used to obtain spatial resolution below the diffraction limit with resolution of 0.09 µm reported[23], however once again it is not implemented inline.

A more salient feature of this technique is that it provides an image of the surface structure, from which specific, local dimensions can be calculated, whereas for the current technology under development, it is desirable to obtain a general overview of the shape and common dimensions repeated in the periodic pattern, especially where the sizes of individual lines can vary over a range.

For this reason, it is more effective to employ a metrology that makes use of the structure periodicity, such as diffractometry or scatterometery, and provides a description of key features, including height, width, sidewall slope and symmetry. Diffractometry is the more attractive option of these two for inline process monitoring since it can provide simple information quickly with less data processing.

Advance beyond the state of the art:

The challenge to progress beyond the state of the art will be to effectively implement diffractometry for inline process monitoring. This will require the design of compact, high numerical aperture optics to capture the maximum number of orders possible.

Similar optical systems have been used to apply sub-wavelength optical diffraction metrology[24],[25] to charactrise structures as small as 50nm in periodic patterns.

As such, VISUM will start by calibrating the system on parts of known geometries with several patterns required for obtaining the effects for each application. Variations of the basic system may be developed, having adjusted light sources and optics to the resolution and spot size needed, and for compatibility with flat surfacesand non flat surfaces. These will be integrated in the related production lines to provide dimensional information on the structures in real time. Resulting data can be used to feed the processing models where appropriate.

[18] http://www.dfm.dtu.dk/index.php?id=309&L=-1%27
[19] Nanometrics Atlas, http://www.nanometrics.com/products/atlas-II.html
[20] Nova T600, http://www.novameasuring.com/t600.html
[21] S.Roy, et al., Coherent Fourier Scatterometry for detection of nanometer-sized particles on a planar substrate surface, Optics Express 22 (2014) 13251
[22] Dr. Schwab DOMS XE-blu, http://www.schwabinspection.com/index/dr.schwab+Inspection+Technology/DOMS+Optical+Disc+V2.html 33 http://www.singulus.com/en/innovative-technology/surface-engineering/crystalline.html
[23] Sensofar S Line, http://www.sensofar.com/metrology/sneox/
[24] Kehoe, Microelectronic Engineering, 86, 1036 (2009)
[25] Silver, Appl. Opt., 46 (20), 4248 (2015). - Kreuzer, Proc. SPIE 9424, 942426 (2015); Kreuzer, Proc. SPIE 9628, 96281Q (2015)-957 (2005)

Selection of processing parameters for LIPSS/Simulation process

State of the art:

As described previously, different types of LIPSS can be obtained depending on the processing parameters. As listed before, the main determinant parameters are the irradiation fluence and its distribution (energy per unit area), number of overlapped pulses, irradiation wavelength, laser polarization and repetition rate. Therefore, the selection of these parameters becomes an essential step in surface functionalization using LIPSS.

Unfortunately, up to now there is not a methodology for the selection of laser parameters in order to obtain LIPSS structures. All the work performed consists of a set of experiments (usually a combination and a variation of different laser parameters like energy, pulse duration, pulse overlap, wavelength, repetition rate) in order to report the relation between two or three of these parameters with the LIPSS structures obtained for a particular material, as illustrated in the next figure[26].

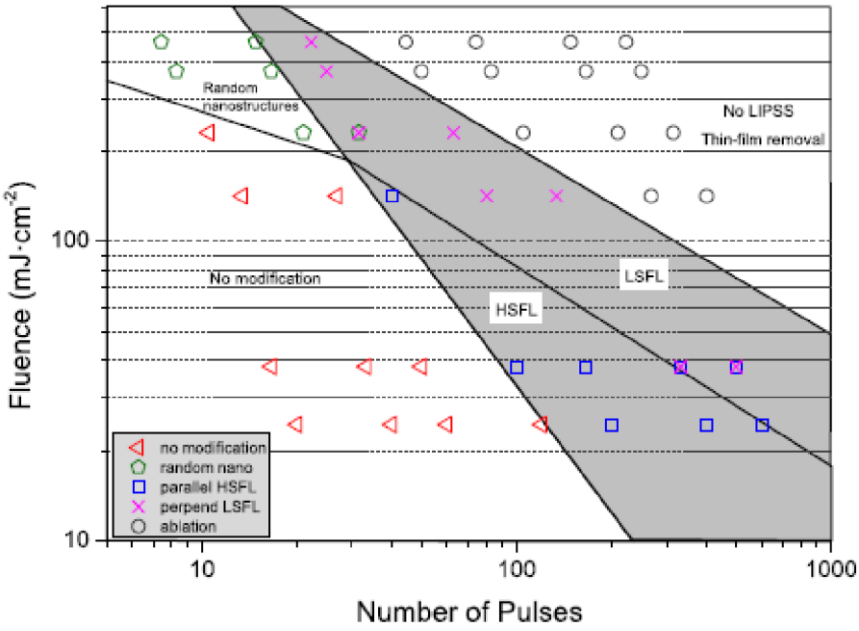


FIGURE 11 TYPES OF RESULTING STRUCTURES AS A FUNCTION OF THE NUMBER OF APPLIED PULSES AND LASER FLUENCE. THE TWO HIGHLIGHTED REGIONS CORRESPOND TO THE SET OF CONDITIONS FOR WHICH LSFL OR HSFL ARE PREDOMINANT, RESPECTIVELY.

For this purpose, researchers usually select a range of values for each of the processing parameters and process material surfaces with combinations of those values/parameters. The ranges for process parameters are selected based on both bibliographic data (mainly for energy determination) and equipment capabilities (scanning speed, repetition rate, etc.), always minimizing processing times.

Although in the last years several research groups have demonstrated the formation of LIPSS on different materials, no tools exist for the inverse problem solution, that is to say, for a specific LIPSS structure which parameters are required. Therefore, there is still a need for modelling tools that enables the selection of the laser parameters that give rise to a specific induced submicron geometry. Additionally, as it has been reported that more than one combination of processing parameters can give rise to the same LIPSS structure, the tool becomes necessary in order to select the optima parameters for maximizing processing time.

Advance beyond the state of the art:

The LASER4SURF project will address the definition of the parameters by developing the necessary tools to predict and optimize the selection of the process parameters. This solution will be based on a statistical analysis of the resulting LIPSS as a function of the processing parameters, in order to establish the correlation between them as well as to determine which parameters become essential for the functionalization process and which are negligible.

LASER SURFACE FUNCTIONALIZATION EQUIPMENT

The three technologies developed in the project will be integrated in a cost-effective Laser Texturing Prototype, for the functionalisation of metal surfaces.

State of the art:

[26] A Rodríguez et al., "Femtosecond Laser-Induced Periodic Surface Nanostructuring of Sputtered Platinum Thin Films", Applied Surface Science 351, 135-139 (2015)

There are some commercial equipment that are able to control the texture in high added value parts using nanosecond and femtosecond lasers currently used in small series manufacturing process of high value components. The more relevant are the following:

	Name	Description	Problems/Limitations	LASER4SURF advance
	AgieCharmilles LASER P 400, GF Machining Solutions	It is equipped with two lasers (nanosecond and femtosecond pulses). It is specifically designed for aesthetic and functional texturing of precision parts for watches and jewelry, small inserts, cutting tools and micro machined work pieces.	▪ Limited to small parts: maximum size of parts is 60 x 40 x 25 cm and Ø 12 x h 12 cm for for 3- and 5-axis systems, respectively. ▪ The distribution of the laser beam is Gaussian ▪ No control of polarization of the beam exists ▪ No in-line monitoring capabilities ▪ No support for parameter selection	▪ Size of parts up to 300 cm x 100 cm x 20 cm. ▪ Beam shaping and polarization control integrated ▪ In-line monitoring capabilities ▪ Support for parameter selection integrated in the machine software
Fraunhofer DLIP systems		Fraunhofer has developed DLIP systems and compact optical heads. The systems can be adapted to the customer wishes in order to fabricate micro or sub-micrometer arrays on large areas. Different pulsed nanosecond laser systems can be integrated into the DLIP systems. Can process both 2D and 3D pieces	▪ Resolution limited to a few microns for metallic materials. ▪ No in-line monitoring capabilities. ▪ Several cm^2 patterned in one shot makes no possible to obtained specific functionalization in precise locations of the sample. ▪ Increased resolution for metallic materials.	▪ In-line monitoring capabilities. ▪ Different functionalizations can be defined in areas as small as a few microns. ▪ Support for parameter selection integrated in the machine software.

DENTAL

LASEA, LS6	It is designed to process large parts such as glass plates and photovoltaic panels. The integration of translation stages and of the laser enable plates to be processed at high speed while keeping high absolute and relative precisions. Different laser sources can be integrated. Shape recognition, autofocus or 2 red pointers crossing at the right focal distance	▪ No in-line monitoring capabilities ▪ No support for parameter selection ▪ The distribution of the laser beam is Gaussian ▪ No control of polarization of the beam exists	▪ Beam shaping and polarization control integrated. ▪ In-line monitoring capabilities. ▪ Support for parameter selection integrated in the machine software.

1.4.1.1 USE CASES

With the aim of enabling the reader to better understand the impact of the LASER4SURF system on the 3 uses cases, an explanation about the current state of the art and the advance proposed is detailed below:

Successful cellular integration is extremely important to the long-term viability of the implant. The sooner the cells can attach and surround the implant, the faster the patient's recovery time, and the lower the incidence of infection and site contamination. The specific biological response of the surrounding tissues enormously depends on the surface characteristics of the particular biomaterial. Cells respond to synthetic micro- and nano-scale topographic motifs in a wide variety of ways, which depend on factors such as cell type, feature size, geometry, as well as the physical-chemical properties of the bulk substrate material. More over mouth infections are currently regarded as the main reason why dental implants fail. Therefore, antibacterial properties are other of the required functionalities to prevent potential bacterial infection.

State of the art: Research groups like CEIT, have studied the cell behaviour in fs-structured surfaces and have obtained very promising results. Cunha et al.[27] found that the fs-texturing of titanium with different nanopatterns can induce the stretching of human mesenchymal stem cells (hMSCs) and potentially improve osteoblastic differentiation. Oberringer et al.[28] observed an improvement of endothelial and bone marrow mesenchymal stem cells on fs nano-textured 316L steel, showing that femtosecond surface texturing could improve osseointegration.

All these findings have been done with no preferential orientation of the nanopatterns. In a very recent work[29], CEIT research group have demonstrated that nanopatterns oriented in a preferential direction, enable and enhance the control of cell adhesion, migration, and polarization without any chemical treatment in a single step process and hMSCs tend to attach and preferentially align to the LIPSS nanopatterns oriented in a longitudinal direction.

Main problems/barriers: These functionalities, has been proved at laboratory environment and need to be developed in order to optimize its performance and reach mass production. The optimal configuration of the oriented nanopatterns along micrometric regions still needs to be defined.

To reach mass production of dental implants, laser texturing technology still need to cover some barriers:

- Very precise dimensional tolerances (~1-2 µm)
- Must not modify the nature of the metal
- Complex geometry to treat – screw threads must remain undamaged ▪ Small size

Advance beyond the state of the art:

LASER4SURF technology will allow to obtain the best configuration for the oriented nanopatterns to improve cell adhesion. The modelling software developed under WP4, will maximise the functionality performance by providing the optimum laser parameters. Moreover, the system database could be updated to cover new materials and improved functionalities.

The multi beam optical module developed in the project, will be able to obtain functionalised metallic surfaces with textures ~1µm or less, enabling the required tolerances to reach the best cell adhesion.

[27] Cunha, A. et al. Nanomedicine (Lond).10, 725–739 (2015).
[28] Oberringer, M. et al. Mater. Sci. Eng. C 33, 901–908 (2013).
[29] Martinez-Calderón el at., Sci. Rep. 2016. http://dx.doi.org/10.1038/srep36296

LINEAR ENCODERS

State of the art: The measurement process of optical linear encoders entails photoelectric sensing of the scale and a scanning reticle that move relative to each other. Projected light is modulated as it either passes through gaps in the gratings when the scale and reticle are aligned, or is blocked when lines of one grating coincide with gaps of the other[30]. The sensor reads the scale to convert the encoded position into an analogue or digital signal, which can then be decoded into position.

In order to have a robust reading, the gratings need to be positioned in a distance equal to the Talbot distance. The Talbot effect is a near-field effect that defines the location of the moving grating of the encoder. This distance is in the order of tens of microns for 4 □m period gratings and must be kept constant with 10% accuracy along the whole scale (as long as 3 meters). To increase the resolution of an encoder, the grating period must be reduced, leading to a decrease of the Talbot distance and thus, complicating the design of the encoder and therefore its cost.

The use of laser treated surfaces has been suggested as an alternative to the current scanning method. The ECOGRAB project works on the design of polarization ratings for its application to linear encoders, as well as a preliminary design of a system for laser processing. The project does not include in-line monitoring capabilities, and neither simulation or modelling tools.

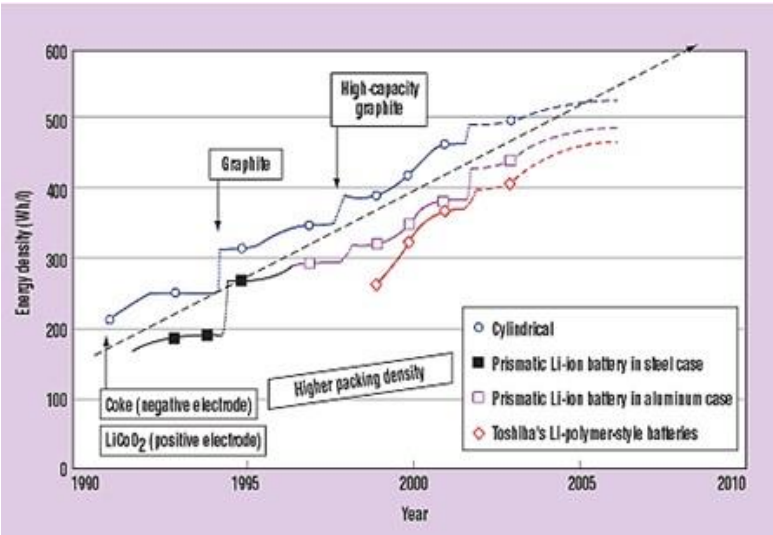
Advance beyond the state of the art: LASER4SURF technology will allow obtaining subwavelength periodic structures in a metal surface, acting as polarizing reflective elements. This polarizing light reflection changes fundamentally the behaviour of Talbot interferences and lifts the need of the accurate positioning, simplifying the mechanical system and enabling a higher resolution of the encoder.

The new surface will change the polarization of the reflected light, which is a novel approximation to optical encoders. The change of polarization in the reflectance is a new effect in the field of encoders, since it introduces novel effects in the near field distribution of the optical interference after the grating. The final result is a cancelation (total or partial) of the Talbot planes, leading to a qualitative increase in the robustness of the encoder system against vibrations and misalignments: the distance between the grating and the optical head with state-ofthe-art technology must be kept constant within a tolerance (for transmittance encoder systems the distance is 200um and the tolerance ±10um, for reflective encoder systems the distance is 1.5mm and the tolerance ±0.15mm). We want these tolerances to increase twofold with this technology.

It will also enable the possibility of higher resolution and a further development of monolithic optical integration of photonic components in glass for optical encoders.

BATTERIES

State of the art: Li-ion batteries are the dominating technology in consumer electronics and power tools. In the past, cell manufacturers have demonstrated that, given a fixed electrochemical couple, energy density of a rechargeable Li-ion battery can be increased by applying new battery designs and packaging technologies.



As demonstrated for other battery technologies,[31] the current collector plays a crucial role in the charge/discharge capacity and cycle stability of Li-ion battery electrodes, for instance, the use of conformal C coatings and etching agents in Al current collectors has been proved to increase the power-delivering

[30] http://www.controleng.com/single-article/motion-control-long-length-linear-encoders-look-lively/70d978580fc5166aca1b94b01e808680.html
[31] Elod Gyenge et al., Current Collector Structure and Methods to Improve the Performance of a Lead-Acid Battery.

capability, by 3–7-fold based on different comparison criteria, of the $LiFePO_4$ commercial electrode, as well as its cycle stability under high C-rate.[32] The increase of current collector surface area by means of foams has also resulted in better battery performance at high C-rates as well as in lower capacity fade. [44] Beyond conventional Li-ion technologies, the current collectors are also a key point for post lithium-ion technologies: three-dimensional current collectors have been used to minimise Li dendrite growth in batteries using Li metal anodes [45]while increasing by 20% the operation time and 30% the average short-circuit time when comparing with planar current collectors.

FIGURE 12: TOSHIBA AND OTHER CELL MANUFACTURERS ARE BOOSTING LI-ION CELL CAPACITY ABOUT 10% ANNUALLY THROUGH IMPROVEMENTS IN PACKAGING TECHNIQUES AND MATERIALS SYSTEMS

Main problems/barriers: The aforementioned methods are based on very complex and expensive structuration and coating techniques. For instance, in negative electrodes, time-consuming (about 50 hours) and complex multi-step processes that involve Cu current collector surface cleaning, deposition of hydroxide coatings by self-assembly methods in ammonia solution based on a two-step chemical reaction, removal of impurities, a dehydration process at 180 °C and a final reduction process in H_2 at 400 °C that will lead to the final structured collector. Whereas for positive electrodes, the Al current collectors are exposed to etching agents that prepare the collector surface for subsequent conformal C-coating by CH_4 heating at 600 °C during 24 hours. However, in 2016 the use of laserinduced micro- and nano-structuration on Cu collectors has been used to improve the adhesion of electrodes; [33] proving that the structuration of Li-ion battery current collectors can lead to a significant improvement of the battery performance. Moreover, the laser structuration can be implemented into the roll-to-roll processes used in the electrode fabrication industry.

Advance beyond the state of the art: Micro- and nano-structuration of current collectors by laser-induced periodic surface structuring (LIPSS) will allow the controlled modification of the current collector surface at a very low cost and very fast (0.1 min/cm^2). The LIPSS process applied to battery current collectors will also involve the improvement of the electrochemical properties without the use of expensive carbon coating methods, harmful etching processes and complex three-dimensional processing methods that have to be used nowadays in the current collectors. The two main advantages of the LIPSS processing will be:

- Absence of chemicals
- Processing speed

1.4.2 INNOVATION POTENTIAL

The LASER4SURF system combines different innovative solutions that cover the whole laser texturing value chain. In the following table the innovation potential of this tools is summarized:

Name		Description	Main innovative characteristics
Complete system	Laser Texturing Prototype	Laser Texturing Prototype based on the definition of Laser Induced Periodic Surface Structures (LIPSS) using ultra-short laser sources, able to texturing 1µm or less structures with precise control of orientation of the sources.	▪ Functionalisation of metal surface by LIPSS. ▪ Improvement of the functionalisation performance. ▪ Processing time of 0.1 min/cm^2. ▪ Cost-effective and up-scalable to mass production. ▪ Integrated inline inspection techniques. ▪ Integrated parameter modelling tool.
	Modelling tool	Software and database that will show automatically the process parameters required to obtain the desired functionality for a specific material.	▪ Transfer the optimum laser parameters to the LIPSS equipment to configure it for the desired pattern. ▪ Enable the end-user to select a specific functionality from a previously defined database. ▪ The database could be updated and enlarged after project end, to cover a wider range of materials and functionalities.

[32] HC.Wu, et al. High-temperature C-coated Al current collector for enhanced power performance of LiFePO4 electrode of Li-ion batteries. Electrochem. Commun. 12, 488–491, 2010.
[44] Yang, G. F., Song, K. Y. & Joo, S. K. A metal foam as a current collector for high power and high capacity lithium iron phosphate batteries. J. Mater. Chem. A 2, 19648–19652 (2014)
[45] Yang, C.-P et al., Accommodating lithium into 3D current collectors with a submicron skeleton towards long-life lithium metal anodes. Nat. Commun. 6, 8058 (2015).
[33] Zheng, Y. et al. Direct laser interference patterning and ultrafast laser-induced micro/nano structuring of current collectors for lithium-ion batteries. SPIE Proceedings 9736, (2016).

Partial solution /individual modules	Optic module	Optic system located between the laser source and the functionalised surface, responsible for the control of the parameters that will determine the LIPSS formation.	▪ Multiple beam processing system (up to 10). ▪ Polarization orientation, fluence and shape control for each beam. ▪ High repetition rate laser source delivering high average power.
	Inspection system	Fast tool for submicron characterization without the need for electron microscopes	▪ Implement diffractometry technology for inline inspection. ▪ Feed the inspection results to the parameter modelling tool.

2. IMPACT

2.1 EXPECTED IMPACTS

According to the FOF Roadmap 2014-2020, manufacturing is a key enabler for Europe's grand societal challenges. The manufacturing sector is in fact the largest segment of Europe's industry in terms of production value and numbers of employees and companies. In order to allow European manufacturing sector to have a significant impact in addressing major social driven targets, an increase in joint government–industry investments in crossdisciplinary manufacturing research is vital, now more than ever, as EU economy calls for **raising the contribution of industry** to **GDP** (Gross Domestic Product) to as much as **20% by 2020**.

HORIZON 2020 is proposing to take action on supporting and promoting business research and innovation in enabling technologies. One of the priority actions will cover advanced manufacturing and processing. Advanced manufacturing is considered by the European Commission as one of the KET (Key Enabling Technologies). The FOF Roadmap highlights the needs of advanced manufacturing processes to produce high added-value functional surfaces to improve product performances and its implementation to mass production conditions.

Surface texturing by direct writing with femtosecond pulse duration lasers is particularly attractive for this purpose because it allows creating a wide range of surface textures, including bimodal roughness distribution textures, in a single-processing step and in a wide range of materials with negligible risk of contamination and thermal degradation of the material

The general impact of the LASER4SURF project is to surpass the technical and economic barriers that nowadays prevent the European manufacturing sector from using laser surface functionalization processes in mass production for metals. By the development of a Laser Texturing prototype, able to texture less than 1µm structures with precise control of orientation.

2.1.1 EXPECTED IMPACTS SET OUT IN THE WORK PROGRAMME

ALIGNMENT WITH EXPECTED IMPACT SET OUT IN THE FOF-06 TOPIC

COST INCREASE PERTAINING TO THOSE FUNCTIONALITIES INTEGRATED INTO PRODUCTS SHOULD BE BELOW 10% WITH RESPECT TO THE COST OF CONVENTIONAL PRODUCTS.

Surface functionalisation by LIPSS, has attracted the industrial attention, due to its main advantages: (i) Flexibility, that enables the modification of a limited area of the manufactured piece. (ii) Versatility, it´s able to apply different functionalities in different areas of the same piece, and to process different materials. (iii) Suitable to work 3D complex pieces.

However, this technology has only been tested in a laboratory environment (TRL 3-4) and local or "niche" applications for short series of high cost products (such as jewellery). Besides, it presents a set of barriers that prevent its use in mass production applications: (i) Process time for the functionalisation of large 3D surfaces, (ii) Long experimental methods to determine the appropriate laser parameters (iii) Extended end-of process inspections techniques.

LASER4SURF will develop an integrated system to tackle these main barriers. For this purpose, the project will address the three main steps concerning the LIPSS process value chain (processing parameter selection, laser functionalisation and inline inspection), and will develop a new technology to reduce process time and cost.

There are not previous initiatives addressing the three main steps of the value chain, for the up-escalation of the process to mass production. Current initiatives focus on process time reduction trough an increase of the power and/or the repetition rate of the laser source. Previous researches carried out by CEIT and MUTITEL have led to scan speeds of 30 min/cm^2 for surface processing of LIPSS on metal surfaces, being the state-of-the-art of laser processing costs **~40 € /cm^2**.

The LASER4SURF project will reduce the current processing cost and time to achieve a cost-effective process, upscalable to mass production, reaching the following figures:

- Total process time of 0.1 min/cm^2. ▪ Cost 0.146 €/ cm^2.

This process cost reduction is sustained in the following cost estimated for the LASER4SURF process:

		Parameters	Production	
Fixed costs	Total machine cost	975.000 €	Hours per day	16
	Years of depreciation	5		
Variable costs	Maintenance	3 €/h		
	Consumables (electricity, gas,…)	0.5 €/h	Annual working days	225
	Operator	30 €/h		

The estimation of the total machine cost is itemized below:

- Machine (280,000 €):
  - Machine LS5-2 with high stability structure in granite and safe compound.
    - Drawer for sampling without production interruption.
  - Air extractor and climatization (electrical cabinet + machine).
  - PLC (Siemens) and HMI for management and automatization.
- Optical modules (40,000 €)
  - Motorized attenuator.
  - Power measurement.
  - Motorized beam expander.
  - Mirrors.
  - Opto-mechanical components.
  - Beam power sampler. ▫ Polarization module.
- Axes: High precision axes system (50,000 €). ▪ Robot module (80,000 €).
  - Robot Staubli (TX60L) for part´s tray loading and unloading.
  - Robot gripper, with automatic robot tool change. ▫ Vision system for the gripper.
- Instruments, vision, sensors (150,000 €)
  - Automatic positioning camera (Cognex)
  - Short range confocal sensor for the profilometry. ▫ Targeting power sensor.
- Automated polarization management module (175,000 €).
- Femtosecond laser (200,000 €).

The process cost has been calculated based on a moderate/conservative approach that estimates a machine working time of 3,600 hours/year (16 hours/day and 225 days/year) and requires an operator during the whole production process, 16 hours/day. However, based on a less restrictive approach (closer to reality taking into account that the proposed solution will be based on an automated machine, where the operator should not be controlling the machine all day) of a 6,000 hours/year working time (24 hours/day and 250 days/year) and an operator working 1 hour/day, it is possible to obtain lower production costs (**0.062 €/cm^2**) than the one used for the calculation of product cost for each one of the use cases (0.146 €/cm^2).

The impact of the implementation of this process on the product cost is demonstrated for the 3 uses cases. These three use cases are a representative sample of the different industrial process where LASER4SURF Prototype could be implemented, as they cover a wide range of different types of materials (Stainless Steel, Aluminium and Titanium) sizes (from 1 cm up to 4 m) and shapes (from plane to helicoidal).

They represent as well a set of three different situations to affront: products **without previous surface functionalities** (like dental screws), development of a **new surface functionality** (like linear encoders), or **improvement of the current product functionality** (batteries) plus adding new ones.

Use case linear encoder:

In this case the use of LIPSS process to treat the linear encoder surface will substitute the actual surface treatment (lithographic process and wet etching for the definition of the grating in the Cr layer) that has a cost of 40€/ unit.

Aspect	AS-IS	TO BE
Annual production	10.000	12000
Dimensions of the product	head 3x6mm ruler up to 3x3000mm	head 3x6mm ruler up to 3x3000mm
Surface Functionalities	Amplitude/Phase control of reflected light	Amplitude/Phase/polarization control of reflected light
Surface manufacturing process	Mask photolithography and wet etching	Laser process
Surface treated (cm^2)	0.18-60	0.18-60
Surface process time	Lithographic process can be complete in a day	0.1 min/cm^2
Cost of the treatment/product	40€	8.76 €
Cost of the product	500€	450€

As the unit cost of the linear encoders is 450 €, the cost increase due to use of Laser Texturing functionalisation will be **1.94 %**. Further cost reduction take place in the mechanical integration of the encoder system as positioning of the gratings is not critical.

Use case implants:

RESCOLL Group's dental implants do not currently use surface texturing.

Aspect	AS-IS	TO BE
Annual production	10 000	50 000
Dimensions of the product	(Several types of screws – ranging from 0.5 cm to 1 cm long)	(Several types of screws – ranging from 0.5 cm to 1 cm long)
Surface Functionalities	None	Laser texturization
Surface manufacturing process	Cleaning & polishing	Laser texturization
Surface treated (cm^2)	1-2	1-2
Surface process time	30 min	0.1 min/cm^2
Cost of the treatment/product	(none, as there is no posttreatment)	0.292 €
Cost of the product	50€ (MSRP)	55€ (MSRP)

As the unit cost of the implant is 50 €, the cost increase due to use of Laser Texturing functionalisation will be **0.5 %.** It is worth to mention that due to the increase properties of the implants it is expected an increased on the annual production (x5).

Batteries:

In this case the LIPSS surface treatment, will substitute the current coating by Carbon Vapor Deposition.

Aspect	AS-IS	TO BE
Dimensions of the product	2032 coin cells	2032 coin cells
Surface Functionalities	Low adhesion, high impedance	High adhesion, low impedance
Surface manufacturing process	Coating (CVD and others)	Laser treatment
Surface treated (cm^2)	The whole current collector roll is treated 30 x 10000 cm	2.5 cm^2.
Surface process time	30 meter/hour	0.1 min/cm^2
Cost of the treatment/product	2.625 €/m	0.35 €
Cost of the product	3 €	3.35 €

As the unit cost of the implant is 3 €, the cost increase due to use of Laser Texturing functionalisation will be **11.6 %.** It is worth to mention that this cost increase will be reduced, as the improvement in electrode adhesion (between 60% and 150%) will help to increase the battery life, reducing the electrode degradation. This improvement in battery life will mean a **reduction of the battery cost in the long term**.

THE IMPROVEMENT IN THE PRODUCT PERFORMANCE SHOULD BE ABOVE 20% IN THE TARGETED FUNCTIONALITIES.

The improvement on product properties obtained by the functionalisation of its surface by LIPSS, has been already demonstrated by the research community. To go a **step further in the current improvement of the product performance** obtained by LIPSS, LASER4SURF project will work on the development of an optical module (to improve the laser performance) and a modelling software (to enhance the laser parameters selections).

As it has been stated in the section 1.4.1 Advance beyond the state of the art section, the irradiation with linearly polarized beams causes LIPSS that are horizontally or perpendicularly aligned to the incident electric field (E-field) vector. The LASER4SURF project will develop an optic module for the Prototype that **will enable the rotation of the laser polarization** with optical retarders, what **will result on the generation of nanopatterns oriented preferentially to a determined specific direction**, and will be able to reduce the size of the patterned structures to less than 1µm, increasing the functionalisation performance.

Moreover, the LASER4SURF project will work on a modelling tool that enables the selection of the laser parameters that give rise to a specific induced submicron geometry.

These developments combined **will improve the product performance above 20 %** as the LASER4SURF project will demonstrate for three different use cases. The product performance increase expected for the three use cases is the following:

Product and target performance (functionalisation)		Current values	Expected values
Batteries	Current rate capability. (Surface friction)	C-coated electrodes have an increase of 7 times with respect to untreated current collector.	An increase between **10 and 30% in the C-rate capability** is expected with respect to the C-coated case.
	Impedance reduction.	5 times reduction of the current collector/electrode interfacial resistance.	Similar values 5 times reduction. **Last indefinitely** since no surface degradation is expected.
		Increase of the resistance with cycling as C-coating is degraded.	
	Electrode adhesion. (Surface friction)	100-150 N/m	>250 N/m, figure based in performed adhesion tests: 90° peeloff **(60%-150% increase)**

Implants	Longer implant lifetime, reduced rejection and better bite strength. (osteointegration and antibacterial)	Torque for explanted screw removal: ▪ 1 month: ~6 N/cm ▪ 3 months: ~25 N/cm	Torque for explanted screw removal: ▪ 1 month: ~7.5 N/cm **(25%)** ▪ 3 months: ~45 N/cm **(80%)**
Encoders	Tolerances of distances between the grating and the optical head. (optical)	For reflective: distance is 1.5mm and the tolerance ±0.15mm.	For reflective: distance is 1.5mm and the tolerance ±0.3mm. **50% increase** in the tolerance
	Incorporating polarization management.	Background light highly variable from part to part	Eliminate the background light.

These improvements on the product performance are based on the following explanation and supported by the state of the art defined previously.

DENTAL IMPLANTS

The improved texturization imparted by the LASER4SURF Project will lead to two benefits: First, the texture will provide simple enhancement of mechanical resistance to removal, by increasing the overall surface area of adhesion. Second, the texturization will also promote enhanced osteointegration with the jaw bone structure, leading to supplementary resistance to removal.

BATTERIES

Current rate capability: The structuration of the current collector surface will increase its effective area and will therefore reduce the electron diffusion length which is the distance that electrons have to travel from the reaction point to the current collector. Depending on the form factor used to modify the current collector surface, it will be possible to increase the effective area by a factor of two when having square periodic features with a height:period ratio of 1:1. In this case the diffusion length will be decreased in average by 50%, leading to a C-rate improvement up to 50%. However, we have considered that several inefficiencies will arise (agglomeration of the electrode material, electrode coating defects, etc) that will bring down the improvement by 50-80% (worst case scenario) leading to an increase in the C-rate of 10 to 30 %

Impedance reduction: The C-coated current collectors have a very good interfacial resistance (5 times better than non-coated collectors). Therefore improving this value with structuration of non-coated collectors will be very difficult. However, increasing the effective area will improve the electrical contact between current collector and electrode. Since the underperformance associated to non-coated collectors is mainly related to the electrode/collector interface, it is expected that structured collectors will have the same impedance as the C-coated ones. The main improvement will appear upon long-term cycling: the C-coated collectors will start failing as the coating is degraded, the non-coated structured collectors will not suffer any degradation problem therefore it is expected that the impedance values remain constant in the long term.

Electrode adhesion: The increase in the effective area will not only improve the electrical contact, also the mechanical one. Having an artificially roughened current collector surface will result in increased penetration of the electrode slurry. This means that the electrode adhesion will be much better than in a flat current collector.

LINEAR ENCODERS

Tolerance of distance between the grating and the optical head.

Optical encoders are formed by two gratings: one in the optical reader head and the other as a reticle. The position between these two gratings must be hold constant at a distance equal to the Talbot distance ZT, that is, the distance at which the gratings produce a self-image by interference. This distance is 〖Z_T=(2a^2)/λ〗_

Being 'a' the grating period and λ the operation wavelength. As the resolution of the encoder increases (period decreases), this distance decreases, and consequently, the mechanical tolerance between gratings is smaller. Therefore, with the standard technology, a compromise between resolution and the cost derived from mechanical restrictions must be reach. The new approach of LASER4SURF, consisting in polarization gratings, introduces a novel effect in near field distribution of the reflected light, resulting in a cancellation of the Talbot effect, and therefore increasing the tolerance to mechanical misalignments of the two gratings.

Eliminate the background light.

Optical contrast is the relation between the minimum and maximum optical signal obtained in the reader head after reflection in the reticle. This signal is always added to a background light that produces a DC offset of the electrical signal. If a polarization grating is used as reticle in encoders, the signal coming from the reticle will present a known polarization direction while the background light will be randomly polarized. Polarization management of these signals will enable the isolation of the background light from the target signal, thus eliminating the background signal and simplifying and making more robust the electrical conditioning of the optical signal.

STRENGTHENED GLOBAL POSITION OF EUROPEAN MANUFACTURING INDUSTRY THROUGH THE INTENSIVE IMPLEMENTATION OF INNOVATIVE AND UNCONVENTIONAL TECHNOLOGIES ALONG THE EUROPEAN MANUFACTURING VALUE CHAIN.

The LASER4SURF project will develop a Laser Texturing Prototype able to be implemented in mass production on the main industrial sectors for the functionalisation of metal and metal alloys based products. The superior characteristics that will be possible to obtain with this system will provide a competitive advantage to European companies, allowing the manufacturing of high added-value products.

Particularly, the implementation of the system proposed in LASER4SURF will have a significant impact on the strengthening of companies of the use cases sectors.

DENTAL IMPLANTS

Statistics provided by the American Association of Oral and Maxillofacial Surgeons show that 69% of adults ages 35 to 44 have lost at least one permanent tooth to an accident, gum disease, a failed root canal or tooth decay. Furthermore, by age 74, 26% of adults have lost all of their permanent teeth. Therefore, the use of dental implants reveals that about 100,000-300,000 dental implants are placed per year, which approximates the numbers of artificial hip and knee joints placed per year.

The dental implant market **currently** has an **increasing number of low-cost competitors** and **no real product differentiation**. Indeed, premium implants are now priced around 230€, while low-cost implants are available for as low as 40€. It is expected that manufacturers of premium implants will not be able to maintain their high margins for much longer, due to the increased pressure from low-cost implants. Therefore, truly **innovative solutions with updated and improved benefits for the customer can help differentiate products with tangible results** and can provide the disruptive influence to drive market growth and profitability.

In this regard, the possibility of obtaining the surface modification proposed in the project will improve cellular adhesion and osteointegration– leading to extended implant life, reduced rejection, and improved overall quality of life for patients. This will provide a competitive advantage to RESCOLL Group against low-cost competitors.

The European dental implants market was valued at 1.9 B€ in 2014 with a CAGR of 9.7%, encompassing the implantation of some 4.5 million dental implants annually. **Europe represents 45% of the global market[47], with the most active markets being the United Kingdom, France, and Germany.**

Indicative average market penetration expected in the next 10 years is based upon 33 implants/10000 inhabitants, leaving a margin of comfortable progression. This translates to sales of approximately 100,000 implants per year in France, Germany, and the United Kingdom. It is expected that Rescoll Group will be able to capture roughly 1.8% of the market through direct implant production and sales to implant vendors.

LINEAR ENCODERS

The European machinery and equipment manufacturing sub-sector is a key enabling sector which boosts Europe's ability to produce new products and services.

With 2.84 million persons employed (equivalent to 2.1 % of those employed in the non-financial business economy and 9.5 % of the manufacturing workforce) and EUR 150.0 billion of value added generated (which was 2.9 % of the non-financial business economy total and 10.9 % of the manufacturing total[34]) is the largest of the 24 manufacturing subsectors, in terms of value added and employment

The machinery and equipment products are characterised by high dynamic operation and accuracy. Position measurements must be provided in tough environments and often under high traverse speeds and accelerations. **Linear encoders with improved accuracy and surface functionalities will help to maintain the European leadership in the sector.**

BATTERIES

Based on the impact results showed above, with the technology developed in LASER4SURF the market of coin batteries can be readily addressed; for larger batteries, more average power of the laser is needed in order to make the process cost efficient. In this regard, new femtosecond lasers of 0.5kW and even 1kW are being developed and should be ready for industrial use within the next years. With the LASER4SURF technology ready, it is expected that a total penetration for the battery market can be made as laser technology evolve.

Latest forecasts by Bloomberg New Energy Finance predict 40 million electric vehicles sold every year by 2040. Considering that those vehicles will be equipped in average with 90 kWh battery packs (2.5 mAh/cm^2 loading of active material in double side coated electrodes), 1200 m^2 of coated current collector will be needed for each battery pack. Two case scenarios can be considered here: one is that only 5% of electric vehicles will mount battery packs with structured current collectors which will involve a considerable amount of structured collector to be manufactured 2400 km^2/year, equivalent to a standard 30 cm wide current collector roll that spins 200 times around the earth; the second scenario considers 100% of electric vehicles using structured current collectors in their battery packs which will mean a fabrication capacity of 48000 km^2/year, equivalent to a standard 30 cm wide current collector roll going to the Sun and back.

Therefore, given the potential of electric vehicle market, the expected increase in batteries capacity and life that will be possible to obtain based on LASER4SURF solution will strengthen the position of EU battery manufacturers, enabling them to compete with Asian manufactures (currents market leaders).

2.1.2 IMPROVING INNOVATION CAPACITY

PROCESS TIME REDUCTION

As it has been mentioned before, one of the main barriers for LIPSS technique to reach mass production, is the high process time required for the functionalisation. LASE4SURF project will work to significantly reduced this process time. To achieve it, the project will

The following time reductions in each one of the main steps of the vale chain are expected:

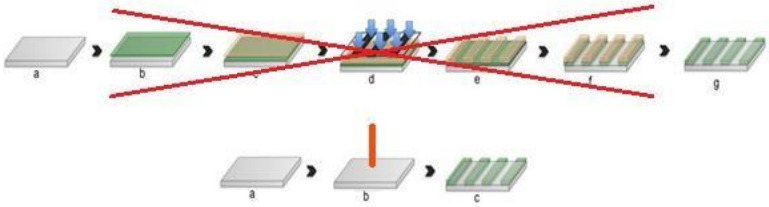
	Modelling	Functionalisation	Inspection
Current	The current process time varies from 13 to 22 h: ▪ 1-lab experiments: 6-10 h ▪ 2-SEM/AFM charact: 6-10 h ▪ 3-Fourier Anal: 1-2 h	30 min/cm^2 < 1 cm^2/min	180 min
LASER4SURF	15 min	1 min/cm^2 10 cm^2/min	1 min

This substantial time reduction of the current LIPSS process, will enable it is up escalation to mass production process. Moreover, this time reduction will contribute to reduce the cost increase induced by the implementation of the LIPSS process to the current products process. To support these the LASER4SURF project will demonstrate

[34] http://ec.europa.eu/eurostat/statistics-explained/index.php/Archive:Manufacture_of_machinery_and_equipment_statistics_-_NACE_Rev._2

the viability of its solution in three for three different products under manufacturing conditions. Use case linear encoder

The LASER4SURF will substitute this multistep process with a single femtosecond laser process. This change will imply a cost reduction that can be understood in terms of a reduction of the equipment used, reduction of chemical needed, reduction of master masks costs and lean fabrication of gratings as needed.



Process		Time
Current process. Photoresist assisted lithographic process and wet etching	Chemical cleaning baths for glass	25 min
	Photoresist spreading technology	25 min
	Cr sputter equipment	180 min
	Insolation equipment	120 min
	Chemical cleaning baths for glass	25 min
LASER4SURF process	Modelling	15 min
	Functionalisation	6 min
	Inspection	1 min

The current process time is 375 min/product, with the new process developed in LASER4SURF the process time per product will be 22 min. Further cost reduction take place in the mechanical integration of the encoder system as positioning of the gratings is not critical

ENVIROMENTAL IMPACT

The use of laser based solutions for the functionalisation of surfaces possess different environmental advantages with respect to other surface treatment methods (chemical treatments, micro-machining…), avoiding the use of raw material and the production of waste and emissions.

- Raw material waste reduction and material reuse.
  Under the the management of dangerous waste, the reduction in the waste generation is considered the most important tool in order to reduce the environmental impact of the industrial processes.
  LASER4SURF techniques permit to avoid the use of raw material for the functionalisation of surfaces as there is now need of adding material for the functionalisation, contributing to a more effective use of resources. Moreover the LIPSS process does not produce any waste material, dangerous for the environmental and health of the operators. Therefore the LASER4SURF project will highly contribute to the reduction of waste of industrial processes as it doesn´t generate waste.
- Substitution of contaminant methods.
  LASER4SURF project will enable the substitution of current contaminant fabrication methods used for the surface functionalisation of products. An example is the substitution of the current method used for linear encoder functionalisation based on chemical photoresists and Chrome thin-films, as well as the current Ccoating made by means of chemical vapour deposition (CVD) techniques.
- Easing the recycling.
  The substitution of the current treatment process, where different raw materials and chemicals are used, for the process developed under the LASER4SURF project, will help to ease the recycling of the treated products, as they will not require the separation of toxics elements from the original product.

SOCIETAL IMPACTS

Employment is one of the most sensitive issues for welfare and living standards for individual income and selfrespect as much as for the European social and tax system. The impact that LASER4SURF contributes to a growth of highly qualified jobs (e.g. process engineers) in the field. It should be noted that the technologies resulting from the

LASER4SURF project do not replace existing jobs with automated systems but only remove existing inefficiencies through new technological solutions in a field that still grows.

The results of the LASER4SURF project will improve the competitiveness of the European industry, favouring a knowledge based industry. This will reinforce the position of the European industry against emerging countries industries. The cooperation between the participants of the consortium, with each one representing a different process of the value chain, will generate new knowledge areas in the laser sector. This will help to raise the added value of the involved companies and research centres, positioning them as technology leaders and contributing to generate direct employment in the consortium participants.

Additionally, the project will have a significant societal impact as allows the improvement in the implants (dental screws, hip and knee replacement…) performance, rising the life time of the implants and easing the recuperation time for the patients.

Every year implants improve the life of millions of patients around the world. The improvement reached by the LASER4SURF project will allow to reduce the two main inconveniences of implants. On one side it will reduce the infection risk between 0,5 % and 5%, the implant infections can imply additional surgical operations. On the other side, it will reduce the recovering time for the patient, thanks to improvement in the osteointegration.

2.1.3 BARRIERS/OBSTACLES

These are some of the main potential barriers, obstacles and framework conditions that can affect the effective implementation of the project results:

Issues	Barriers/obstacle	Mitigation plan
Reproducibility	The global added tolerances of all components might be different from a machine to another, i.e; beam size, divergence, waists positions are hardly the same from one laser source to another, and can lead to a global set of characteristics out of range.	To put in place accurate and systematic checks, the characteristics of the machine, and make possible some adjustments to get back into tolerances ranges specs. This includes the use or the development of specific instruments.
Robustness	The first machines integrating such complex optical components, will have a high percentage of breakdowns and needs of maintenance, with non-negligible costs and be time consuming.	Put in place a strict follow-up and bug reporting, to take into consideration in order to improve the efficiency of the fabrication process for every following machine.

Besides, it is important to mention that the following barriers/obstacles have been identified regarding the implementation of the new functionalised products that could be possible to obtain thanks to LASER4SURF solution:

Issues	Barriers/obstacle	Mitigation plan
Regulations	**Implant:** Delay due to regulatory approval for implants	**Implants:** approval in the dental implant market is much faster than other markets (up to five times faster), hence its use as a first market.
Market	**Dental Implant:** severe competition	RMG will be able to produce and sell the implants (provided the price impact is 10% or less) at the 50€ price point. This price range is currently occupied by “bare bones” implants lacking high-end functions like surface texturing – typically observed only in the 200€+ price range. This will make the implants highly competitive.
Others	Batteries: Difficulties to significantly alter the existing production battery lines to integrate the LIPSS processing.	Batteries: Focus on new production lines under development.

2.2 MEASURES TO MAXIMISE IMPACT

Communication, dissemination and exploitation are the three pillars for creating an **appropriate strategy to maximize market uptake of the project results** involving the entire supply and value network for the project development, implementation, knowledge sharing and future exploitation of the results. The optimal and prompt utilisation of project results by relevant stakeholders requires an effective dissemination and exploitation strategy relying on:

- A clear identification of all **Target Groups** (Industrial Community, Research Community, Policy-makers and regulators, platforms and information portals and general public)
- **Tailor made communication tools and according dissemination actions** for both the identified players of the value chain and the general public.
- The development of a **comprehensive Exploitation Plan** including: Business plans (tested and validated by potential end-users) for the main project results and recommendations for market uptake of each.
- **Efficient IP management** in order to facilitate rapid commercial exploitation
- An efficient **Data management system**

2.2.1 DISSEMINATION AND EXPLOITATION OF RESULTS

2.2.1.1 DRAFT PLAN FOR THE DISSEMINATION AND EXPLOITATION OF THE PROJECT'S RESULTS

DISSEMINATION PLAN

LASER4SURF is a market-oriented project and its success will be very dependent on how the project will reach the targeted stakeholders, so that they are informed about the projects outputs and the potential of the obtained results. The following targeted stakeholders and key messages have been pre-identified:

Target group	Goal	Type of information
Industrial Community: manufacturing companies, potential end-users, Laser platform's stakeholders	Share experiences, mobilised sector interest by demonstrating the added value, innovation and new technologies developed within the project continue R&D activities, market technology	▪ Experiences reports ▪ Business cases
Research Community (universities, research centres and R&D divisions companies related to the laser sector), including clustering with other European R&I projects and H2020 initiatives;	Raise awareness, spread knowledge between researchers and students about the technology	▪ Original scientific findings and results ▪ Experiences reports
Policy-makers and regulators: the consortium will strive to address the policy level for supporting the ultimate exploitation of the project results.	Influence policy priorities	▪ Market potential evaluation ▪ Technical criteria and scientific findings
Platforms and information portals: youris.com, sciencedaily.com, horizonmagazine.eu, www.photonics.com, www.osa-opn.org	Awareness	Examples of use cases
General Public: citizen/consumer organisations	Boost the knowledge and possibility of laser for metal surface functionalisation.	Examples of use cases.

In order to carry out the project dissemination in a comprehensive way, as a first step, a **Communication Master Plan** will be produced in month 3 of the project and will be updated in month 20. The updated version of the Plan will take into account a report on activities already implemented (also to be published in M20) and give an outlook for the dissemination activities foreseen for the next period. This document will set: headline project information and key messages by audience segment; identification and classification of main target audiences, and

communication channels; project branding guidelines: visual and written identity; implementation plan and timeline; Roles and responsibilities of partners. Besides, it will establish a network of dissemination and communication ambassadors and spokespersons within the project. A summary of **Communication tools and Dissemination channels** is included below (more detailed information is included in section 2.3:

Communication tools and Dissemination channels

Dissemination will be achieved thanks to several communication tools adapted by the project and the use of appropriate dissemination channels:

- **Visual identity:** a project logo, a reference PowerPoint presentation, templates for deliverables and general documents will be designed in the very initial phase of the project;
- **Written content:** Twelve interviews and six journalistic articles, which take an independent view on the project, and provide a wider view on the issue of surface texturing. These articles are to be disseminated for publication on different online publishing sites like sciencedaily.com, youris.com or similar.
- **Visual content:** Six Info-graphics and two brochures will be produced during the course of the project to support online and offline dissemination
- **Audio-visual content:** two videos will provide a gateway to the project and used in a number of different presentations of the project and its exploitation.
- **Scientific peer-reviewed articles (Open Access),** contributing to dissemination of knowledge generated and Open Science and Responsible Research & Innovation (RRI) policies;
- **Online dissemination:** By accessing **web multipliers** and a **social media campaign,** LASER4SURF will build up a community that includes project posts on relevant and existing social networks, contributing to raise awareness and keep attention toward project activities and achievements high; For this activity, all the original content (written, visual and audio-visual) will be used.
- **Offline dissemination**: Printed communication materials such as brochures will support the communication with direct 1:1 contact with different stakeholders. In addition, **workshops and field visits** will be organised for providing in-depth knowledge of the technologies used in the project to potential end-users and for first-hand knowledge sharing.

The project deliberately avoids using **newsletters** and **press releases** as a form of communication to stakeholders, as they have proven largely unsuccessful in the age of information overflow. Instead the LASER4SURF will focus on the production of compelling and attractive content that can be fed into different dissemination channels. Working with web multipliers, social media and engaging in direct 1:1 dialogues with stakeholders, the impact will be much larger than with the use of periodic newsletters or press releases.

The following table summarizes the main tools used to disseminate and communicate the project progress and/or results to the respective target groups:

Dissemination and Communication Channel	Industrial Community	Research Community	Policy Makers & Regulators	General public
Web Site	X	X	X	X
Advisory Group	X	X		
Links with Associations	X	X	X	
Scientific conferences / journals		X		
Trade fairs / publications	X		X	
Specialist media	X	X	X	
General mass media				X
External Project Events	X	X		
Networking / Clustering Events	X	X		
Online Dissemination Material (incl. social media)	X	X	X	X
Offline Dissemination Material	X	X	X	

Deliverables	X	X	X

ESCI will be the main partner for the dissemination activities and will maintain a schedule of the dissemination activities that will be undertaken by the Consortium and will promote within the Consortium specific actions that should be targeted by several participants. The dissemination strategy, achievements and plans will be delivered in specific reports. All partners will actively collaborate and contribute to dissemination activities to optimize dissemination actions towards commercial exploitation. Therefore, they shall also use their own dissemination networks to further publicise the Project and thus ensure maximum visibility and impact at a regional, national and EU level.

EXPLOTATION PLAN

Beyond measures to disseminate project outcomes, specific tasks leading up to the finalisation of business plans for each key result and the development of an overall **Exploitation Plan** is foreseen. The laser developer LASEA will act as manager of the exploitation activities and ensure that all IP-owning partners as well as potential and confirmed exploitation partners will work towards a successful technology and knowledge transfer for speedy market uptake.

Identification of the exploitable products and services will be a continuous process during the project. At this stage, according to the project objectives and the expected outcomes, the key project results that can be exploitable are the following: Laser texturing prototype (Global system), optical module, Monitoring system and modelling software.

Exploitation issues will be brought to the attention of all partners from very early in the project to ensure that findings regarding market and commercialisation that come up when studying exploitation options can still be fed into the R&D part of the project.

EXPLOITATION PLANS OF THE PARTNERS ON THE ACHIEVED RESULTS:

Below is a description of the main exploitation routes and strategies for each partner:

	Specific EXPLOITATION ROUTES and STRATEGIES
LASEA	LASEA will enrich its product portfolio, thanks to new designs of machine integrating new components that allows LASEA to achieve new performances. **Surface functionalization**: Besides the significant improvement of the functionality of the machines thanks to the control of the different modules, LASEA is also interested in the improvement of its knowledge regarding surface functionalization using the LIPSS. The control of the functionalization can open the way to new markets of applications, for example applications in tribology, control of the friction, anti-icing, aesthetics, …
CEIT	CEIT is a RTO that works very close to industrial partners offering them the development of laser processes for improving their products. After project ending, these industrial partners usually require CEIT for the fabrication of pre-series of their new products with the developed processes. With the actual technology, CEIT cannot offer them an affordable solution. However, once LASER4SURF technology is available, by acquiring a new machine Ceit will be able to offer a complete solution to its industrial partners. Additionally, the results of LASER4SURF will be used by CEIT in order to identify and develop further applications for other markets such as automotive, home appliances or food Moreover, the in-line monitoring system developed by VISUM-IRIS will be a very fast tool for submicron characterization without the need for electron microscopes. This tools could be attached to other CEIT's equipment or even operate in a stand-alone mode.
MULTITEL	Besides the main development of MULTITEL in this project, i.e. the dynamic beam and polarization control module, other results are of interest for MULTITEL from scientific and technical point of view: **LIPSS materials texturation:** MULTITEL already proposes today micromachining services to companies and the project LASER4SURF is an opportunity for MULTITEL to learn more about the LIPSS processing and potentially include this technology in its service offer. **Online control and monitoring**: As a laser and laser processes developer, MULTITEL is also very interested by the questions related to process online control and automatic feedback on laser parameters. This is an opportunity to develop more adapted and flexible laser sources for operating with such modules that will, in the future, be part of most of the laser micromachining systems. **High energy ultrashort (picosecond) lasers:** This is a third and side part of the project LASER4SURF that is part of the core business of MULTITEL. The test of MULTITEL own lasers during the project for process development and validation of the beam shaping and polarization control modules will be an added-value for MULTITEL from exploitation and dissemination point of view.

VISUM	The activities performed under LASER4SURF will directly impact competitiveness of VISUM as hardware, software and communication module and will create return on investment as well as will enhance product competencies and knowledge of condition monitoring applications into other industrial sectors like food and pharma which are strategic for the company. Technology transfer: all knowledge acquired during the project will be integrated into VISUM consulting services. This will concern new real-time monitoring techniques, advanced control strategies, adaptive and intelligent post-processing methods, data fusion of non-destructive inspection methods. **In-line Optical monitoring system:** This module would be the main system that VISUM would be keen on finding the right exploitable route among the user cases in order to transfer this knowhow to other sectors which Optical technologies are positioned.
CIC	The LIPSS structuration of current collectors is the main subject in the energy validation part. However, there are other interesting results: **LIPSS structuration of other battery components**. The possibility of structuring other battery components using LIPSS opens many research possibilities that could lead to further improvements in the battery performance. **Application of LIPSS processing to other battery technologies** such as Na-ion, Li-S, Li-metal and solid state batteries. The structuration in other battery technologies could induce new functionalisations that overcome some of their associated drawbacks.
FAGOR	FAGOR will internally exploit LASER4SURF results, directly in their production lines. The incorporation of LASER4SURF technology to their production lines will not only reduce the processing time for the fabrication of encoders but also the characterization time, that is currently performed off-line in a separated line.
RESCOLL	RESCOLL Group's primary exploitation of the results in this project will be on dental implants. However, the dental implants will be used as a **proof of concept for other markets** in which the company is actively involved, including **spinal screws, knee and hip implants, and shoulder implants**. Due to the fact that regulatory restrictions are far more severe for these markets, the information from the approval for the dental implants will be used as a foundation for building the dossiers for other applications later on. Whereas dental implants could see market introduction in as early as 12 to 18 months, the delay for other implants is often at least 60 months.

SYNERGIES WITH OTHER FOUNDS

Public funding: The redesign procedures of funding programmes at a European level has resulted in more harmonized instruments, specifically: Horizon 2020, COSME and the European Structural and Investment Funds (ESIF). This, combined with the possibility to use different public funding sources within a programme, project or group of projects (combined funding), has enhanced the synergies between those funding instruments and others defined at national, regional levels and even private funds for ambitious industrial projects.

Private funding: After the project completion, LASER4SURF partners will co-develop the final products with customers to adapt results of the project to specific needs and get industrial readiness level. Financing of these tasks will be based on internal funding by the partners, but they will also look for new product and service development funding at national and regional level, as well as ESIF funding and others.

2.2.2 BUSINESS PLAN

The final solution of the project is envisioned to be marketed through LASEA. Other partners will be compensated with the payment of licenses and/or royalties for the services and improvements that contribute to the solution but will not have a variable compensation based on unit sold.

Additionally, the rest of developer partners (CEIT, MULT, VISUM) will market their developed solutions individually (modelling tool, optic module and inspection system). A preliminary summary of the exploitation plans of each one of the exploitable results is included in the following tables. The following exploitable results have been identified.

Exploitation Result 1: Global system	
Main Exploiter	LASEA
Main Innovation Concept	Thanks to the modules that will be developed within this project, LASEA will be able to offer laser systems with improved capabilities for their application for the functionalization of components by laser. The textures that we will be able to functionalize thanks to the laser system that we could provide integrating the developed modules will be more reproducible, of better quality, and generated automatically. The data base should help the use of the laser system in choosing the best parameters for his application.

Main identified customers		At a first level the companies end-users and research centres, participating in this project. Thanks to the flexibility characterizing in general the laser technology, we expect as well to be able to offer similar solutions to companies in other areas of application.
Market and its size		It's expected to be able to get customers, first, by geographical proximity in Europe but as LASEA is a company working all around the globe, we expect to be able to commercialize results Worldwide.
Positioning	**Value Proposition**	▪ Minimization of operation time. ▪ Minimization of cost increase. ▪ Maximization of functionalization performance.
	Competitors	Other engineering companies, like for example TRUMPF, ROFIN, 3D MICROMAC or OXFORD LASERS.
	Value for outbidding competition	By relation between price and quality. LASER4SURF will provide a unique laser texturing system with characteristics not currently offered by any company.
	Protection vs competitors	Some results could be patented and this will be identified during the project development thanks to the task related to the exploitation and IPR aspects.
Time to market		As the project is aiming to TRL6, it will be necessary to work further on the project results in order to achieve a higher TRL. In principle, before commercialization should be necessary to develop a prototype of laser system, possibly different depending on each application. This could demand between one and two extra years after the project completion.
Roadmap		LASEA will try to start commercialization with the end users already involved in this project, trying to arrive to agreement afterwards in order to produce their expected laser system for their application.
Economic impact in the company		At least an increase in the turnover in the first 5 years after the completion of this project of around 4 million euros based on an estimation of the sales of at least 6 machines. At least 3 different machines for the end-users already participating in this project and 3 others that will be found in different applications.
Issues to be solved		The global system includes developments of different partners. Therefore, special **during the** exploitation agreements to organizes the IPR of the different parts participating in the **project/Challenges** foreground, of the global solution, generated in the consortium will be discussed.

Exploitation Result 2: optical module	
Main Exploiter	MULTITEL
Main Innovation Concept	System is based on a milliJoule femtosecond IR source coupled to high speed galvanometric and an online control of the polarization orientation and beam profile. The module for polarization and beam control will be a stand-alone unit that can be adapted to different type of lasers and machines.
Main identified customers	LASEA is for instance one potential end-user for such technology but any other laser integrator and machine producer could be a potential client for this module.
Market and its size	The market of laser processing in industry is worldwide. This is an increasing market, as we can see from the table below showing the market evolution of ultrashort lasers in industrial applications:

Table 3.2 Material Processing Ultrafast Laser Summary 2014 to 2019

Materials Proc	2014	2015	2016	2017	2018	2019	CAGR 2015 to 2019
Femtosecond							
Units	482	623	767	930	1,106	1,295	20.1%
Prices ($)	$285,000	$256,000	$237,000	$221,000	$208,000	$199,000	-6.1%
Revenue ($ Million)	$137.5	$159.5	$181.8	$205.4	$230.1	$257.7	12.7%
Picosecond							
Units	1,075	1,142	1,572	1,946	2,486	2,969	27.0%
Prices ($)	$162,000	$136,800	$115,200	$108,000	$97,200	$93,600	-9.1%
Revenue ($ Million)	$174.2	$156.2	$181.2	$210.1	$241.7	$277.9	15.5%
Total							
Units	1,557	1,764	2,340	2,875	3,592	4,264	24.7%
Prices ($)	$200,094	$178,881	$155,134	$144,531	$131,317	$125,608	-8.5%
Revenue ($ Million)	$311.6	$315.6	$362.9	$415.6	$471.7	$535.6	14.1%

Source: Strategies Unlimited

In industrial applications, each unit represents a laser with a machine. This machine will include most of the time a beam shaping element. In very few cases a dynamic beam shaping with control of polarization will be required but, as we see with the three application cases of this project, such demands also exist. With the evolution of technologies (SLM, diffractive optics, introduction of LIPSS process,...) the demand for such "exotic" features will continue to increase.

Positioning	**Value Proposition**	▪ Implementation of a spatial light modulator based polarization and beam shaper for producing controlled polarized beams: linearly, circular, azimuthally and radially polarized beams. ▪ Implementation of multi-beam splitting functions
	Competitors	Companies like PULSAR PHOTONICS, AdIOptica (Germany), QIOVA (France) for instance propose beam shaping solutions based on SLMs.
	Value for outbidding competition	To our knowledge none of them propose online control of polarization and we believe that a simple system including polarization rotation and control combined with diffractive optics will be more cost effective and robust to high power and high energy applications.
	Protection vs competitors	Patent will be considered as an option during the project.
Time to market		After the project completion, more packaging, long term reliability and stability tests will be necessary. If the results of the project are already sufficiently advanced (at a TRL 6), 2 years more will be sufficient for really reaching the market.

Roadmap In first place, we will need to work with early technology adopters like the company LASEA with who we could establish specific but non-exclusive agreements. Joint publications and presentation of the product in specific fairs like LASER WORLD of PHOTONICS Europe, Asia and PHOTONICS WEST will permit to reach a large community of potential integrators. The aim will be to participate to these fairs for presenting the potential product with a booth

and poster or oral communications. Besides, the future product will be firstly experimented with LASEA during and after the project but other potential early adopters will be contacted (through for instance the ECOLASERFACT database). Ideally the system should be under test at three different premises (can be industrial or research laboratory with worldwide reputation) one year after the project termination. From each test, a communication will be prepared to the industrial and scientific community.

Issues to be solved during the project	The main issue from the project will be to clearly define the most interesting potential product from the real application cases. Technologies allow for various options and approaches that are not always the most adapted solutions from economical and application perspectives. The fact that LASER4SURF includes all the value chain is a valuable advantage to properly fit the technology to the real application needs and integration constrains.

Exploitation Result 3: Monitoring system		
Main Exploiter		VISUM
Main Innovation Concept		In-line Optical scatterometry monitoring system
Main identified customers		Metallurgical Industries , Steel Industries and any kind of industrial companies that would be using metal with Nano textures (Automotive Industry, Aviation, Home appliances and even medical sector)
Positioning	**Value Proposition**	Monitoring solution able to in-line and on-line acquire the size and the data coming from the laser texturing process will be developed and integrated in the processing head. This monitoring system will be capable of detecting anomalies of the texturing process and subsequently feed that information back to the process control stage for corrective measures to be applied.
	Competitors	Engineering companies and RTDs that offer non-destructive test and metrology services
	Value for outbidding competition	Nowadays all monitor systems requires a lot of time in preparation of the samples and just few user cases are able to apply the system in-line.
	Protection vs competitors	Patenting the monitor module with the specific configuration of the optics
Time to market		2 years after the project end
Roadmap		Once the project is finished, the results obtained will be transfer as application note under the commercial portfolio of VISUM. After 6 months, there will be published several services related to the monitor system and the overall network of partners and customers will be able to test the performance of this technology under their premises. Finally, a year later, it is expected that there will be the first services provided to industrial partners.
Issues to be solved during the project		The main issue would be related to adapt the monitor system to each user case according to the non-linear samples. Since the range of the test under the LASER4SURF would be quite wide, the tuning of the optical parameters will be key.

Exploitation Result 4: parameter selection software		
Main Exploiter		CEIT
Main Innovation Concept		The software to determine the optimum process parameters for a selected material, application and functionalisation.
Main identified customers		Laser equipment integrators and machine tool companies, R&D centres, industrials partners of CEIT in material development, additive manufacturing, automotive, railroad and aerospace industries and requiring the incorporation of laser processes to their production lines .
Market and its size		While most of the current customers are located in north Spain, the dissemination and exploitation of the results of laser4surf will address all relevant present and future customers of Ceit at a EU level
Positioning	**Value Proposition**	▪ Software tool supporting the selection of laser parameters required for specific functionalization of metallic surfaces ▪ Upgradable Database for future processes
	Competitors	Other RTO that can offer the development of specific surface functionalization processes
	Value for outbidding competition	To our knowledge none of them propose any kind of support for parameter selection or any affordable update of the developed ones

	Protection vs competitors	Patent and/or industrial protection will be considered as an option during the project.
Time to market		2 years
Roadmap CEIT will start contacting its actual customers already interested in laser texturing solutions, although it will also start contacts with new customers in the potential replication sectors identified in WP7		
Issues to be solved during the project		Identification of opportunities in other markets (WP7) A need to address the right companies in the different markets value change and compatibilism of the customers from different markets.

Additionally, the Laser prototype developed under the LASER4SURF project will be tested in three different use cases. A preliminary Business Plan for the improved products related to the uses cases is included in the following tables. These Business Plans will be defined in detail during the project.

Use case: Linear encoders		
Main Exploiter		FAGOR
Main Innovation Concept		Optical encoders more robust to misalignments and simplified electrical conditioning of the optical signal
Main identified customers		Machine Tool Manufacturers
Market and its size Optical Encoder Global Market was valued at 1.300 M€ in 2015 and it is expected to expand at a CAGR of 8.7 % during the period (2016–2026)		
Positioning	**Value Proposition**	▪ Optical encoders more robust to misalignments ▪ Elimination of background light to simplify electrical conditioning of the optical signal
	Competitors	HEIDENHAIN (mainly), others (Lika Electronic Encoders, Sony, Anilam Magnetic Encoders, Acu-Rite, Ono Sokki, RSF Probes & Glass Scales, Northstar Tachometers, Metronics, Newall, Solartron Metrology)
	Value for outbidding competition	To our knowledge, none of the potential competitors uses polarization gratings as optical encoders
	Protection vs competitors	Patenting will be considered during the project. Additionally, the specific shape and distribution of LIPSS on metals make them difficult to replicate by potential competitors not owning the same fabrication techniques
Time to market		If the results of the project are already sufficiently advanced, 3 years more will be needed to really reach the market.
Roadmap Once the project has finished and the functionalization process has been defined, the industrialization of the process will be necessary. Additionally, an optical head should be designed and fabricated for the new optical encoder.		
Economic impact in the company		If the project objectives are fulfilled, a 10% increment in the company's annual turnover is expected, once the exposed market is available to the company. Exposed linear encoders are designed for use on machines and installations that require especially high accuracy of the measured value and are not accessible for conventional linear encoders
Issues to be solved The functionality of the developed processes must be validated and, for that purpose, an **during the** specific testing setup must be design and developed at Fagor facilities. **project/Challenges**		

Use case: Dental implants

Main Exploiter		RESCOLL Group
Main Innovation Concept		Medical implants with improved osteointegration and antibacterial capacities.
Main identified customers		The improvements of the LASER4SURF project will help RESCOLL Group to position itself in the bottom end of the "premium budget" market.
Market and its size		Medical implants have a wide market divided in various products. The use of dental implants reveals that about 100,000-300,000 dental implants are placed per year.
Positioning	**Value Proposition**	▪ Improved osteointegration ▪ Antibacterial capacity ▪ Reduced recovery time for patients
	Competitors	Low cost product manufacturers from countries like India.
	Value for outbidding competition	Improve the implants performance in terms of mechanical resistance of the integrated implant, the reduced recovery time of the patient, shorter medical treatment, reduced gingival problems probability in the long term Also for the European manufacturer to have a technical step forward in comparison to the, e.g. Indian low-cost implants.
Time to market		For dental implants 12-18 months. For other medical implants (knee, shoulder…) could be up to 60 months due to regulatory restrictions.
Roadmap		Once the project has finished and the functionalization process has been defined, the industrialization of the process will be necessary.
Economic impact in the company		The improvement of the dental implants surface due to the functionalisation by LIPSS will turn in an improvement of the product sales turning in a 50% increment in the company's annual turnover, once the exposed market is available to the company.
Issues to be solved during the project/Challenges		The final production cost of the surface treatment process should be low enough to enable RESCOLL Group to achieve, the movement from one of their main clients of the entire products rage to LASER4SURF technology.

Use case: Batteries		
Main Exploiter		CEGASA
Main Innovation Concept		Structured current collector for high power rechargeable batteries.
Main identified customers		Electrode manufacturers Custom Cells (Germany) Johnson Matthey Battery Materials (UK) LiFeSIZE (Sweden)
Market and its size		The main markets are niche applications as military and medical applications as well as a percentage of stationary and automotive applications. In 2016 the rechargeable lithium-ion market was €23.000M and these applications are a 15% of the market.
	Value Proposition	▪ Increase of the power density ▪ Increase in the lifetime ▪ Negligible cost increase

Positioning	**Competitors**	Manufacturers of C-coated current collectors Armor Group (France) Coveris Advanced Coatings (UK) Showa Denko K.K. (Japan) Toyo Aluminium K.K. (Japan) Vacuum Carbon Technologies (Russia)
	Value for outbidding competition	Higher quality, longer life
	Protection vs competitors	Industrial secret
Time to market		5 years

Issues to be solved Implementation at industrial scale keeping high production rates **during the project/Challenges**

2.2.3 HOW THE PARTICIPANTS WILL MANAGE THE RESEARCH DATA GENERATED AND/OR COLLECTED

LASER4SURF project will join the **Open Research Data Pilot** (initiative which aims to improve and maximise access to and re-use of research data generated by projects). Partners will provide open access to research data only in the case of public Deliverables, non-protected results or results coming after a patent registration, so that the exploitation is guaranteed and the commercial interest of results is not jeopardized.

- A **Data Management Plan** (DMP) will be set up on month 6 and will be updated during the project spam life on an annual basis. The DMP aims to provide an analysis of the main elements of the data management policy that will be used by the applicants with regard to all the datasets that will be generated by the project and provide a solid procedure to guarantee open and free access to them. Main contents of the DMP will be:
- Definition of **dataset units** (Scientific publications; compilation data tables…) to be produced during the project life including: origin (in case it is collected), nature and scale and other relevant information. The definition will also include, in case a dataset cannot be shared, the reasons behind (e.g. ethical, rules of personal data, intellectual property, commercial, privacy-related, security-related).
- Definition of a **procedure to standardize the metadata publication format**, as described in the grant agreement: Unless it is impossible, any communication activity related to the action and any infrastructure, equipment and major results funded by the grant will display the EU emblem and include the following text: (i)For communication activities: “This project has received funding from the European Union’s Horizon 2020 research and innovation programme grant agreement No [number]”; (ii) For infrastructure, equipment and major results: “This [equipment] is part of a project that has received funding from the European Union’s Horizon 2020 research and innovation programme under grant agreement No [number]”.
  When displayed together with the LASER4SURF logo, the EU emblem will have appropriate prominence.
- **Data sharing**: Description of how data will be shared: i.e. A selection of some specific Bibliographic social networks such as http://www.citeulike.org, http://www.mendeley.com or http://www.bibsonomy.org were all presentations, contributions and publication even partially funded by the project will uploaded no later than six months after its original date of publication.
- Description of the procedures that will be put in place for **long-term preservation** of the data.
  To assure proper Data Management, appropriate specific points to regulate ownership and access to key knowledge (IPR, data etc.) are included in the Consortium Agreement already signed among all the partners.

2.2.4 STRATEGY FOR KNOWLEDGE MANAGEMENT AND PROTECTION

The **strategy for knowledge management and protection** (IPR Management Plan) is setup in the Consortium Agreement to be signed by the consortium and included in the WP8 deliverable LASER4SURF Exploitation Plan. This will include partner’s already existing knowledge, rules to distribute new IPR created within the project and

protection measures to assure proper exploitation of innovative results. The overall IPR strategy of the project is to ensure that partners are free to benefit from their complementarities and to fully exploit their market position. Hence, the project has a policy of patenting where possible.

LASER4SURF proposes a complete range of activities leading to the optimal visibility of the project and its results, increasing the likelihood of market uptake of the knowledge it produces, and ensuring a smooth handling of the individual intellectual property rights of the involved partners in view of paving the way to knowledge transfer. **Internal knowledge management** will be facilitated through a web-based secure professional collaborative space (such as 'Teambox') for information and document sharing. Each partner will have a section where he can upload key documents and papers.

Regarding **ownership of the foreground**, it is agreed that foreground will be the property of the beneficiary carrying out the work generating that foreground. Where several beneficiaries have jointly carried out work generating foreground (e.g. Framework, Services) and where their respective share of the work cannot be ascertained, they will have joint ownership of such foreground. They will establish an agreement regarding the allocation and terms of exercising the joint ownership, including definition of the conditions (compensation to other joint owner(s), time plans etc) for granting licenses to third parties. Conditions of the transfer of ownership of the own foreground to 3rd parties is predefined in Consortium Agreement ensuring that the rights of the other Project Partners will not be affected by such transfer. Beneficiaries will also define the **background needed** for the purposes of the project development in a written agreement and, where appropriate, may agree to exclude specific background. **Access rights to foreground and background** will be granted to the other beneficiaries, if it is needed to enable those beneficiaries to carry out their own work under the project and/or to use their own foreground provided that the beneficiary concerned is entitled to grant them. Such access rights shall be free of any administrative transfer costs and will be granted under fair and reasonable conditions or on a royalty-free basis unless otherwise agreed by all beneficiaries in a specific agreement. All access rights will be granted upon written request.

For use of project results in **research and teaching without any commercial character** the respective project partners receive access rights on royalty free basis. The Parties undertake to cooperate to allow the timely submission, examination, publication and defence of any dissertation or thesis for a degree which includes their Foreground or Background. The concerned project partners receive access rights to the project results under reasonable conditions. LASER4SURF will assure the open access to **scientific publications** relating to foreground published before or after the final report details/ references and an abstract of the publication must be provided to the Commission at the latest two months following publication. Furthermore, an electronic copy of the published version or the final manuscript accepted for publication shall also be provided to the Commission at the same time for the purpose set out in Article II.12(2) if this does not infringe any rights of third parties.

2.3 COMMUNICATION ACTIVITIES

The aim of the communication activities is to build a greater awareness and enthusiasm for laser surface texturing and all communication and dissemination activities will be produced and implemented in view of a successful exploitation strategy.

First of all, a **logo and payoff** with the acknowledgement of EU funding and EU flag will be developed to create an identity that will be specific for LASER4SURF actions and partners, and to distinguish the project from other projects, in line with H2020 visual guidelines. LASER4SURF branding will be used on publications and for project dissemination and communication.

Online dissemination is at the centre of the communication strategy, as it allows the consortium to reach a variety of key target groups, spread the word about LASER4SURF actions and technologies and thus maximise their impact. Online dissemination provides the best method to monitor the dissemination success, measure its effect and enables an adjustment if necessary.

- The **website** is a reference point and anchor for LASER4SURF online content and outreach activities. It will explain the context, developments and ambitions of the project to our stakeholders and the general public. In order to keep a continuous and current information flow from the project to the public at large, the website will contain articles about LASER4SURF topics, interviews with experts from within and outside the project, and short news reports about project highlights.
- To facilitate collaboration and productivity for the consortium, a **protected space** will feature appropriate working documents, reporting and non-public deliverables.

- Joining the dots with social media actions, the **LASER4SURF twitter feed** will be prominent on the home page and well referenced across the site. For the twitter feed, the grouping will identify stakeholders and influencers, building lists to help strategic and geographic segmentation with the aim to attract and maintain the interest of key influencers and thought leaders. On this channel, LASER4SURF will distribute original content and enhance and amplify presence before, during and after workshops and other events (see Subtasks 8.2.6 and 8.2.7)
- In the professional domain, **LinkedIn** will be used to host a 'company page' to feed with project news and developments and targets invited to follow. The platform's 106 million unique monthly visitors will also generate healthy organic search and reference for LASER4SURF content, with an additional possibility to use the sites 'pulse' publishing features. Rather than try to create an LASER4SURF group from zero, content will be placed where a highly interested and well-populated specialist community already exists.
- Further to these, **youtube** will be used host the videos and LASER4SURF editorial content will be pushed to a number of **websites for science, technology and specialist media**. These web multipliers include sites such as ScienceDaily and AlphaGallieo or Optics & Photonics where highly interested professionals and media source a lot of their content for further use and distribution to their own audiences and followers.

A variety of **editorial and visual content** will be developed to share on LASER4SURF channels, media multipliers and championed by influencers on social media as well as printed dissemination materials to support face to face meetings, at workshops, fairs or conferences. The following tools and actions will be implemented:

- **Editorial, visual and audio visual content:**

  Written content: six original journalistic articles, which take an independent view on the project, and provide a wider view on the issue of surface texturing will be commissioned and published on the project website, and be shared with influential multiplier websites in specialist media, stakeholder networks and on platforms like LinkedIn. The articles will also be pitched to local, national or international mass media in the ESCI network. In addition, **12 interviews** will focus on key technical experts, end users and stakeholders who will relate to their experiences, ambitions and challenges in achieving interoperability, optimisation and demand responsive innovations. These interviews will be published on the project website and their quotes and points of view will drive social media activities and promotion.

  - **Visual content:** In order to communicate complex information effectively and in an eye-catching format six **info-graphics** will be developed to demonstrate key concepts, interoperability issues, best practices and results. They are highly useful in for social media communication, presentations, postcard flyers and posters. An initial series of three info-graphics will outline the key concepts for the laser technology LIPSS, the in-line inspection process and provide the context how specific industries (e.g. dental screws or battery production) could benefit from these tools. A further three will be produced towards the end of the project to profile key results and give clear calls to action.
  - **Audio-visual content:** two videos will provide a gateway to the project First, a short, concise and creative **introductory video** using animation and clear call to actions, will aim to capture and keep the attention of target audiences – inciting their curiosity and driving them to stay informed about LASER4SURF outputs. This first video will be produced in M4 and will serve as a key feature of the website and for repeated exploitation at external events and various social media (twitter, facebook, youtube). The **final video** will target stakeholders at large so that they can identify, understand and begin to apply the results of the project. The final video is to become an important tool for the exploitation of the project's results and will be presented at the final conference in M36 and be integrated on the project's website and the various social media outlets.
- **Printed communication and dissemination material: postcards** will be produced based on the info-graphics developed to demonstrate the most important elements of LASER4SURF. In addition, in M6 a project's **brochure** will be designed in order to provide additional information on printed material for fairs and conferences for all partners to distribute individually to potential end-users and other stakeholders. This brochure will also be updated when tangible results of the project become available. Besides, an **A1 poster** giving the key facts and outputs of the project will be designed and made available for all consortium members to produce and use themselves.
- **Publications (articles and specialized articles):** LASER4SURF has the potential to generate articles peer-reviewed scientific papers, submitted to open-access scientific journals and magazines in the field of laser and laser metal surface texturing as well as technical papers or poster presentations submitted to highly respected conferences

in the relevant fields. The following are already foreseen: Applied Surface Science, ACS Applied Materials & Interfaces, Advanced Materials Interfaces, Journal of Power Sources, NanoLetters.

- **Events and workshops** will be arranged and organised for providing in-depth knowledge of the technologies used in the project to potential end-users and for first-hand knowledge sharing. In this regard, a number of communication actions have been designed, using the communication material describe above. The following events have already been identified (The final list will be defined in WP8): Photonics West, LASYS, World of Photonics (Lasers in Manufacturing), LAMP, AILU workshops, ICALEO, JNPLI, International Meeting on Lithium Batteries, International Conference on Advanced Lithium Batteries for Automobile Applications

To ensure optimum outreach, the organisation of three events has been planned (back to back or in the context of other larger relevant sector-specific events, to attract a high number of stakeholders. The scope of those workshps will be laser applications and advanced laser technology.

The consortium has excellent access to clusters & networks through already existing and well established contacts. They are part of a number of international associations, that can multiply the dissemination of LASER4SURF content, for example: Photonics 21. The European Technology Platform Photonics21 has more than 2500 members. CEITik4 is member of the Board of Stakeholders (BoS) of Photonics 21; SECPhO (Southern European Cluster in Photonics and Optics) brings together companies, technology centres and research groups in the photonics and optics sector. CEIT-ik4 is member of the Board of Directors. Secpho has 65 members.

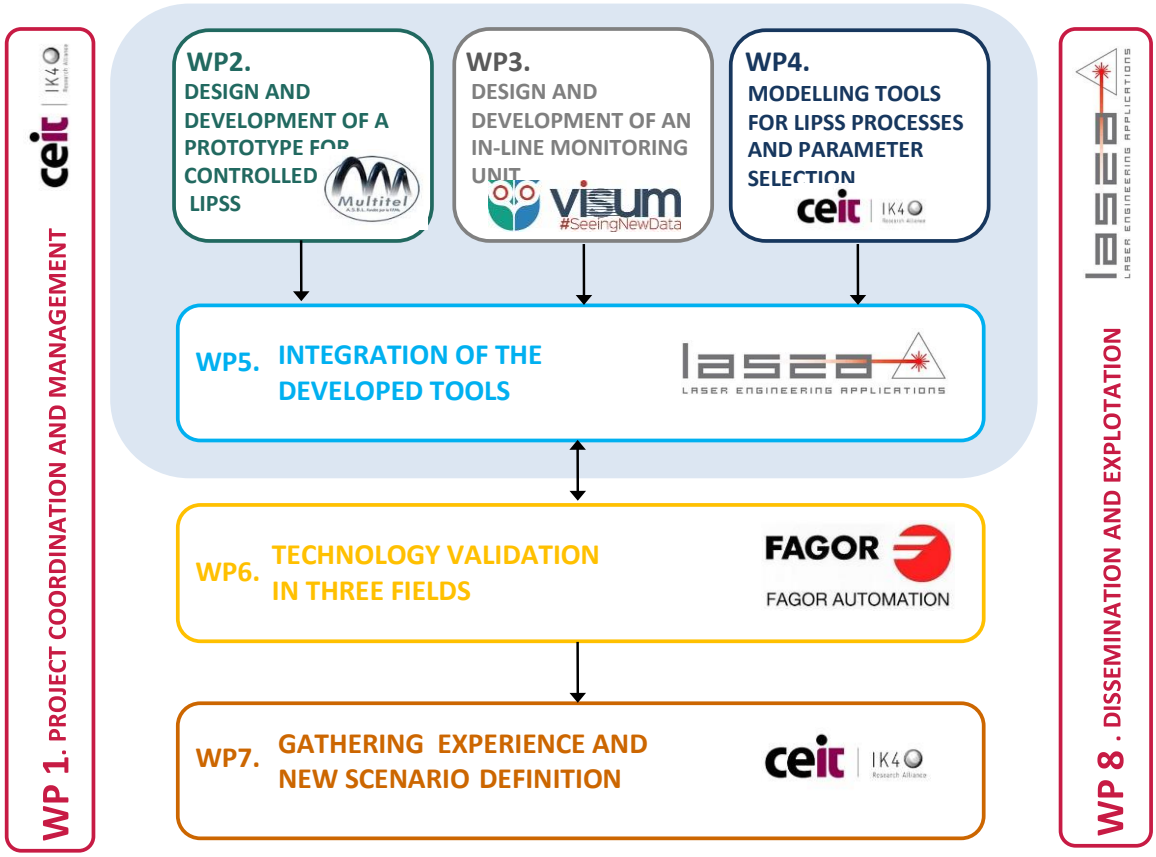
3. IMPLEMENTATION

3.1 WORK PLAN

3.1.1 BRIEF PRESENTATION OF THE OVERALL STRUCTURE OF THE WORK PLAN

To achieve LASER4SURF´s overarching goal and the specific objectives, LASER4SURF work plan is divided into eight Work Packages. Besides the cross-cutting Work Packages, i.e. WP 1, dedicated to project management and WP8 (Dissemination and exploitation) there are three main blocks of Work Packages:

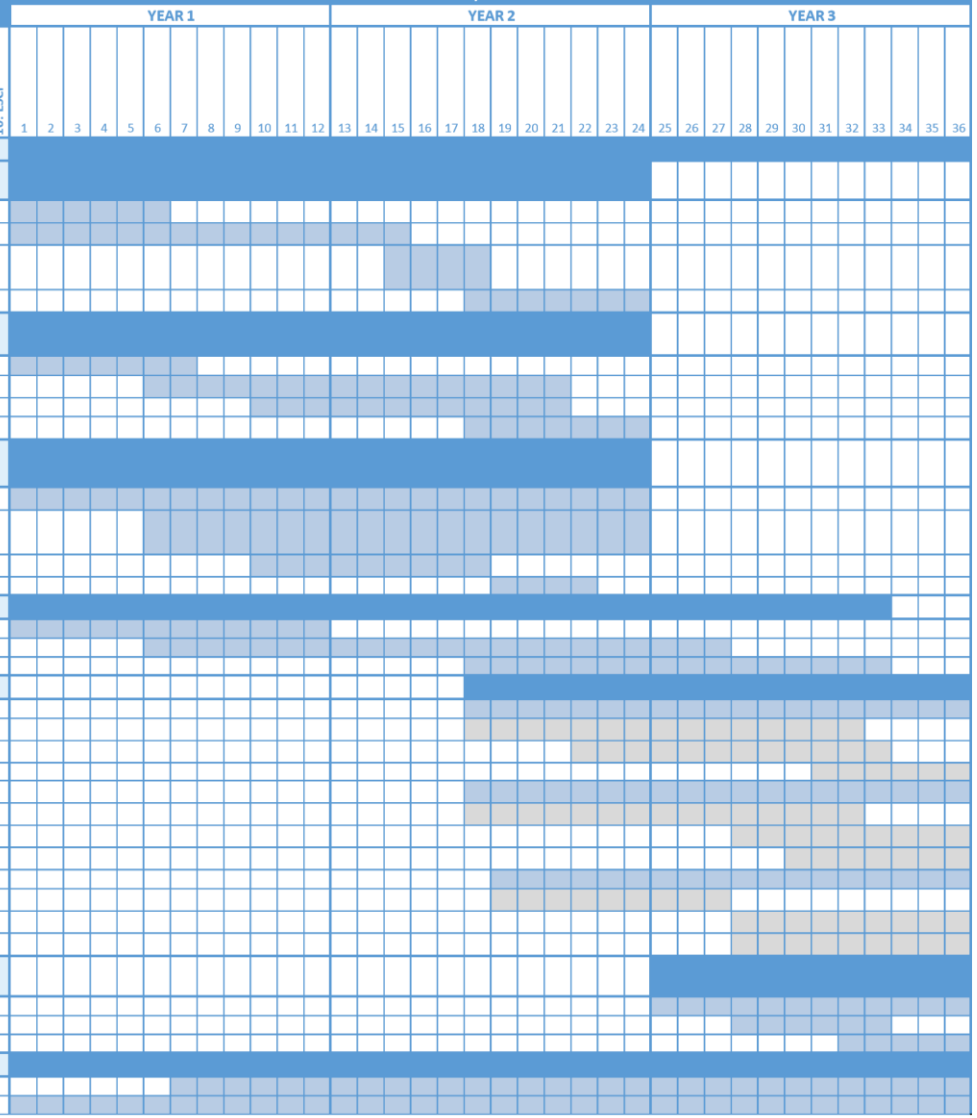
- The first block (WP2, WP3, WP4 and WP5) is dedicated to the development of each module of the Laser Texturing prototype and its integration.
- The second block (WP6) is dedicated to demonstrate the feasibility of the proposed laser texturing prototype in a set of 3 products of three different sectors: Machine tool industry, healthcare sector and batteries.
- The third block (WP7) addresses the project´s replication. With the aim of maximising the future impact of the project results, this WP will define new strategies for an effective replication of the process developed, not only in the specific sectors covered by the use cases, but also in other sectors.



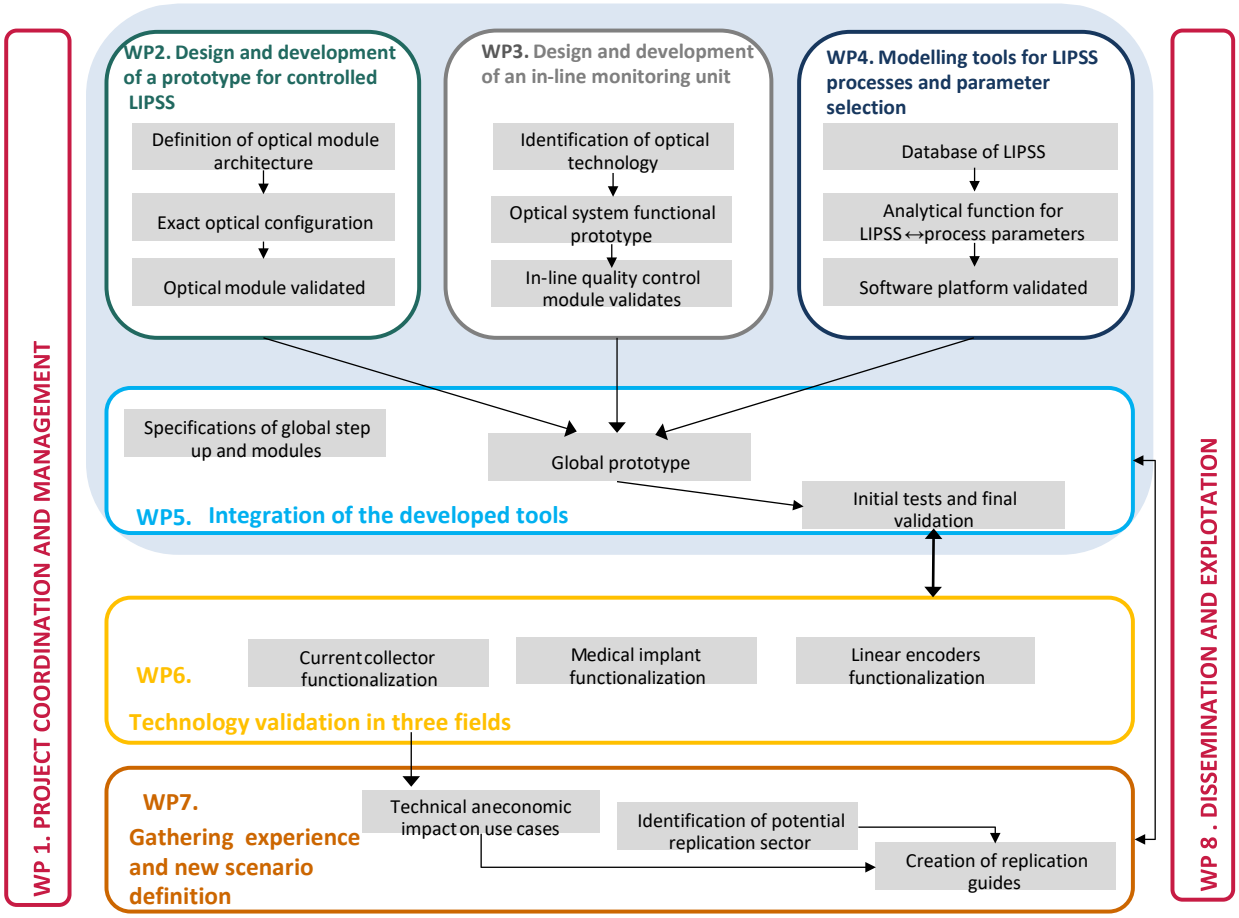
3.1.2 TIMING OF THE DIFFERENT WORK PACKAGES AND THEIR COMPONENTS

WORKPACKAGE DESCRIPTIONS	EFFORT (PMs)									
	1. CEIT	2. LASEA	3. MULTITEL	4. VISUM	5. Iris (TP)	6. CICe	7. Cegasa (TP)	8. FAGOR	9. RESCOLL	10. ESCI
WP1 PROJECT MANAGEMENT	L	P	P		P	P	P	P	P	P
WP2 Design and Development of a Prototype for Controlled LIPSS	P	P	L	P		P		P	P	
Task 2.1 Project Requirement Analysis and Definition of the System Architecture	P	P	L	P		P		P	P	
Task 2.2 Preparation of the experimental laboratory set-up and tests	P	P	L							
Task 2.3 Definition of the optical, electrical and mechanical interfaces of the optical modules	P	L	P							
Task 2.4 Realisation and test of the optical modules	P	P	L							
WP3 DESIGN AND DEVELOPMENT OF AN IN-LINE MONITORING UNIT	P	P		L	P			P	P	
Task 3.1 Core optical system development	P	P		L						
Task 3.2 Data acquisition and processing software development	P			P	L					
Task 3.3 Functional prototype development		P		P	L					
Task 3.4 Nano-metrology lab testing and validation	P	P		L						
WP4 MODELLING FOR TOOLS FOR LIPSS PROCESES AND PARAMETER SELECTION	L	P	P			P		P	P	
Task 4.1 Generation of a database of LIPSS processes and resulting functionalities	L	P	P			P		P	P	
Task 4.2 Statistical analysis of the influence of process parameters on the obtained LIPSS and functionalities	L									
Task 4.3 Design and development of a software platform for parameter selection	L	P	P							
Task 4.4 Pre-validation of the software tool	L	P	P							
WP5 INTEGRATION OF THE DEVELOPED TOOLS	P	L	P	P	P		P	P	P	
Task 5.1 Study of the integration	P	L	P	P	P		P	P	P	
Task 5.2 Integration of the diffrent modules	P	L	P	P	P					
Task 5.3 Test and validation	P	L	P	P	P					
WP6 TECHNOLOGY VALIDATION IN THREE FIELDS	P	P				P	P	L	P	
Task 6.1 Technology Validation in Energy Storage.	P	P				L	P			
Subtask 6.1.1 Fabrication of structured current collectors for pouch cell prototypes	P	P				L	P			
Subtask 6.1.2 Monolayer pouch cell fabrication and electrochemical characterization.	P	P				L	P			
Subtask 6.1.3 Characterisation of pouch cell electrodes	P	P				L	P			
Task 6.2 Technology Validation in Biomedical Implants		P							L	
Subtask 6.2.1 Fabrication and dimensional characterization of modified implants		P							L	
Subtask 6.2.2 Mechanical and chemical analysis and validation of the modified implants									L	
Subtask 6.2.3 Biological validation of the added functionality									L	
Task 6.3 Technology Validation in Linear Encoders	P	P						L		
Subtask 6.3.1 Design and Implementation of a validation setup								L		
Subtask 6.3.2 Fabrication and dimensional/optical characterization of diffraction gratings	P	L						P		
Subtask 6.3.3 *Validation of the fabricated gratings in optical encoders*	P							L		
WP7 GATHERING EXPERIENCE AND NEW SCENARIO DEFINITION	L	P	P			P	P	P	P	
Task 7.1 Use case analysis	L	P	P			P	P	P	P	
Task 7.2 Identification of potential replication sectors	P	L					P	P	P	
Task 7.3 Replication guides	L	P	P	P						
WP8 DISSEMINATION AND EXPLOITATION	P	L	P	P	P	P	P	P	P	P
Task 8.1 Exploitation	P	L	P	P	P	P	P	P	P	
Task 8.2 Communication & Dissemination	P	P	P	P	P	P		P	P	L

DURATION / CRITICAL PATH



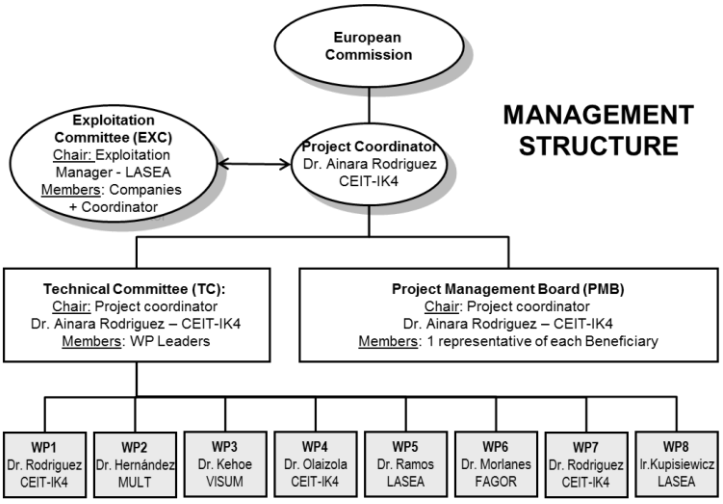
3.1.3 GRAPHICAL PRESENTATION OF THE COMPONENTS SHOWING HOW THEY INTER-RELATE



3.2 MANAGEMENT STRUCTURE AND PROCEDURES

3.2.1 DESCRIBE THE ORGANISATIONAL STRUCTURE AND THE DECISION-MAKING

Project management structure – In order to avoid management difficulties, the LASER4SURF consortium has put great care in agreeing and defining a simple leading hierarchy that ensures fast decision-taking mechanism and a smooth project management while providing the necessary control and participation mechanisms.



The management structure has been organized in two main boards chaired by the Project Coordinator:

- The **Project Management Board** (PMB), composed of one representative of each partner of the consortium.
- The **Project Technical Committee** (TC), composed of the WPs leaders.

Apart from these 2 main bodies, other smaller bodies have also been defined in the project:

- The **Work Package Teams**, one per WP, composed of representatives of the partners participating in the Work Packages. These WP teams are leaded by the WP Leaders.
- The **Exploitation Committee** (EXC), composed by the Exploitation Manager, Project Coordinator and representatives of the different companies in the consortium

Project Coordinator - The project will be coordinated by CEIT, being the project coordinator Dr. Ainara Rodriguez.

She received her degree in Telecom Engineering from the University of the Basque Country in 2006 and her PhD from the University of Navarra in 2010. She has a wide experience in the participation and coordination of R&D projects related to laser material processing. She is co-author in more than 40 peer-reviewed papers and communications in conferences. She is also co-applicant in 3 patents. Dr. Ainara Rodriguez has considerable experience and a track record managing and coordinating multi-partner collaborations in Spain and the EU.

Moreover, the project coordinator will be supported by the resources of the Materials and Manufacturing division and Commercial Department of CEIT-IK4, which have a vast experience in FP project management. These two departments comprise the necessary skills and profiles, including expertise in managerial issues (highly experienced project managers specialized in EU FPs), technical experts, experts in legal issues (lawyers with experience in previous FP7 Grant Agreements and H2020 Grant Agreements, IPR issues, contract management aspects etc.), and financial experts specialised in R&I funding.

The Project Coordinator and her team will be responsible for: (i) Monitoring that the action is implemented properly. (ii) Organising, steering, checking and assuring the reporting in time towards the European Commission. (iii) Quality assurance of the documents submitted by the beneficiaries. (iv) Submitting the deliverables and reports. (iv) The coordinator is responsible for all communications with the Commission. Act as the intermediary between the beneficiaries and the Commission. (v) Monitoring of the expenses and allocation of the budget, assisting the participants concerning administrative aspects of the project; (vi) Coordinating the payments to the partners. (vii) Requesting the needed amendments if necessary. (viii) Providing assistance towards the participants concerning administrative, financial and legal aspects of the project.

Project Management Board (PMB). A Project Management Board as steering and management committee will be set up. It will be chaired by the project coordinator and it will comprise representatives of each of the partners taking part in the project. One main aim of this Board is to advise and support the decisions of the Project Coordinator on operational and management issues. This board will be responsible for all decisions of general nature within the frame of the EC Contract and the Consortium Agreement. The PMB will be responsible for the “major decisions” affecting the implementation and success of the project such as: (i) Any decision affecting more than one partner such as contractual matters, planning, financial matters, major technical decisions, preparation of reporting. (ii) Major changes in the nature of the project including starting or stopping it to conduct a particular part of the project. (iii) Any expenditure out of the agreed in the budget. (iv) Issues concerning the ownership of the results and access rights of the results. It is foreseen that the PMB will meet every 12 months in face to face consortium meetings and whenever needed in audio conferences. The meetings will be organised by the Coordinator.

Project Technical Committee (TC) It will be convened every 6 months. The TC will exchange the work carried out in the different WP and clusters in order to support the implementation and running of the project. The Project Technical Committee will play a crucial coordination role among partners for the implementation of the different case value chains.

Work Packages Teams and Leaders - The WP Leaders will assure the coordination between the different project teams that collaborate with the aim of exchanging intermediate results. They will assure the timely execution of tasks included in each WP, stimulating the interaction between the various partners involved. They are also responsible for the consolidation of the specification reports

and execution of the tasks that integrate each WP. Each organization involved will appoint a WP Manager, who is responsible for operational decisions.

Exploitation committee (EXC) - The EXC will be chaired by LASEA. The aim of EXC is to enable the optimization of the exploitation possibilities of the project results and to cover all the aspects related to future exploitation and IPRs issues. It will comprise the consortium industrial partners, Project Coordinator and those partners in charge of the development of the modules. The whole value chain will be represented in this exploitation committee. It will be convened every 12 months for the two first years and every 6 months during the last 18 months. The EXC will play an important role for decisions regarding exploitation of results.

3.2.2 ORGANISATIONAL STRUCTURE AND DECISION-MAKING MECHANISMS ARE APPROPRIATE TO THE COMPLEXITY AND SCALE OF THE PROJECT

General Management Procedures - Partners are responsible for: (i) Effective economic management and conduct of the operational work in accordance with the program guidelines and with ethical and legal standards. (ii) Complying with the general terms and conditions governing grants and any terms and conditions specific to each grant or granting program established by the European Commission. (iii) Managing and supervising operational personnel. (iv) Meeting reporting requirements specific to the H2020 and the specificity of the call. (v) Acknowledging, whenever possible, the Coordinator's financial support for the operational work.

Internal communication flow – LASER4SURF is a project that involves different partners with different fields of expertise that will work together to develop and validate a laser texturing prototype that will allow to merge the definition of process parameters + functionalisation + inspection in one single process. In order to ensure collaboration, fulfilment of goals and good general functioning of the consortium, there will be 6 monthly meetings (TC) meetings that will include representatives from the WP Leaders. This will guarantee the synchronization of all the tasks, the definition of control procedures to follow the evolution of work, solving potential conflicts between partners, defining dissemination policies for results and planning the presentation of common communications. Teleconferences, email, ftp and web exchange will be used as the primary forms of communication and exchange of documents among the partners. The Consortium will hold annual full partner meetings to disseminate technical progress through the presentation of results, to peer review and to coordinate and ensure coherence to project planning. Consensus in decision-making is highly desirable, both for quality assurance and to ensure that the goals and interests of the partners are respected.

Monitoring and progress reporting - Each WP leader will report project progress to the Project Coordinator on a six-monthly basis. This will cover technical progress, results, deliverables and compliance with the WP schedule, as well as the monitoring and updating the identified risks.

Quality assurance - To assure the quality of the project, a Project Management Handbook will be elaborated at the beginning of the project by CEIT, summarizing all the information required for an optimum management of the project (in terms of administrative forms, procedures, templates, financial aspects, communication procedures, deliverable schedules, reporting periods, contacts, etc.), and conflicts resolution. Additionally, the project will apply an internal reviewing procedure to guarantee the quality of its results. Each WP leader will be responsible for the quality of the results, especially deliverables. These will be subjected to a peer review by at least one expert: another WP leader – the one who will use the results of the WP being reviewed as input.

Conflict Resolution - Conflicts will be solved at the lowest level possible, and preferably amicably. If an agreement cannot be reached at a task or WP level, then the Project Coordinator will mediate. Procedures will be provided in the CA to be signed by the partners before the signature of the grant.

IPR Management- Intellectual Property Right Issues and the exploitation plans are directly under the control of the PMB. The state of the required existing knowledge and the rights to use the developed knowledge will be the subject of specific IPR annexes to the CA that will be developed in line with the recommendations of the EC, following the rules and regulation of the EC Model Contract. The commercial exploitation of the results will take place in conformity with the interests of the EU and in accordance with the aims of the Horizon 2020 programme objectives. In addition, the CA will formalise the basis for the use of background material and other IPR issues (i.e. product results exploitation, results usage rights within the consortium…).

3.2.3 HOW EFFECTIVE INNOVATION MANAGEMENT WILL BE ADDRESSED IN THE MANAGEMENT STRUCTURE AND WORK PLAN

The management structure of LASER4SURF has been specifically designed to be an effective innovation management system, taking into account the three following main aspects:

- Organizational structure: the innovation requires organization at different stages, coordinating level, operational level and support level; that is the reason why a Project Coordinator, a Project Technical Committee, a Project Management Board, Work Packages Leaders and an Exploitation committee have been designated.
- Internal communication flow: Having a good communication within the consortium is one of the key elements of innovation management and decision making. Thus, the chairpersons of the different boards have been specifically designated to allow a flow communication between partners; they have the necessary communication and management skills and also flexibility to be capable of adapting to unexpected changes during the project.
- The definition of the work plan is essential for achieving an effective innovation management system. During the proposal preparation all the partners have jointly defined the Work Packages, its objectives and expected results, paying special attention to Dissemination and Exploitation tasks in order to achieve market objectives.

3.2.4 CRITICAL RISKS

The management process will identify and monitor technical and management risks as well as any other issues that might affect the project progress towards its objectives, in order to carry out mitigation actions as early as possible Risks can arise from: i) Unexpected technical difficulties or scientific findings; ii) Poor communication or cooperation between the partners; iii) Resource shortage by the partners; iv) Objectives not achievable in terms of budget or feasibility; v) Human operational errors: planning errors, poor quality, incomplete tasks, etc.

According to the risk analysis performed by the partners during proposal preparation, there are some technical/management critical risks relating to project implementation.

3.3 CONSORTIUM AS A WHOLE

3.3.1 CONSORTIUM DESCRIPTION

The **LASER4SURF** consortium is composed of 8 organizations from 5 different countries (Spain, France, Ireland, Belgium and Germany), bringing together the necessary competence and critical mass in terms of expertise and resources to **ensure the achievement of the project goals and the impact of its results at European level.**

The consortium involves a well-balanced group of highly qualified entities with different backgrounds and complementary expertise and skills. Partners have been **chosen to cover the whole value chain**, constituting an interdisciplinary group of experienced and consolidated partners, each of them providing its expertise to cover the different fields required by the project: from modelling of LIPSS parameters, laser texturing process and inline monitoring to validation simulating operational environment conditions as well as exploitation and dissemination of the results. A brief description of each partner and the main role in the project is shown in the next table:

	PROFILE	MAIN ROLE IN THE PROJECT
CEIT	**RTD partner.** CEIT will participate in LASER4SURF project through the Optoelectronic Microsystems research. The use of laser systems to process the material surface for a range of applications, including solar cells, semiconductor lasers and biomaterials is one of the two main activity areas of the group.	**Main Role: developer of LIPSS process database and software tool developer. Project Coordinator** (WP1). CEIT will **lead WP4**, focused on the development of the software platform that will allow the selection of the process parameters that give rise to a desired functionality, without a need to deeply understand the LIPSS generation process. CEIT will also contribute in the definition of the requirements of the prototype, especially in the design of the optical module (**WP2**) and in the integration of the developed tools (**WP5**). CEIT will be actively involved in **WP7** to maximise the future impact of the project results and in the dissemination & exploitation tasks (**WP8**).
LASEA	**SME**. LASEA has a unique expertise in the design, manufacture and use of laser systems for industrial purposes. LASEA offers cutting edge laser solutions for companies whose production lines include plastic welding, marking, and surface treatments. LASEA has manufactured laser production lines all over the world	**Main Role: Integrator of global system.** LASEA has years of experience in laser processing for industrial purposes. Therefore, LASEA will play a key role in the development of the project. It will **lead** the integration of the modules in the global prototype (**WP5**). Besides, due to its vast knowledge of the market, it will be responsible for **WP8**, leading particularly the exploitation activities.

MULTITEL	**RTD partner.** Its activities include signal processing, wireless networks, railway certification, image processing and applied photonics. The Applied Photonics Department of MULTITEL has developed an expertise in use of ultrashort pulse lasers for high precision processing of materials	**Main Role: optical module developer.** Due to its vast experience in ultrashort pulse lasers, in LASER4SURF project, MULTITEL will be addressing the beam shaping issues, **leading WP2**. Besides, it will collaborate in the integration of the optical module into the processing unit developed in WP5.
VISUM	**SME.** VISUM is an optical engineering company specializes in the development and commercialization of Near Infrared Spectroscopy (NIRS)-based portable and inline analyzers and measurement systems for process monitoring and quality control applications.	**Main Role: in-line monitoring developer.** The optical engineering and smart systems teams at VISUM has specialized expertise in sensor integration, optical engineering, chemometrics, data analytics, cloud infrastructures and data management systems. Thus, VISUM will be **responsible for** the Design and development of the in-line monitoring unit (**WP3**), defining the optical technology suitable for nanometric surface features quality control. Besides, it will participate in the integration of the developed in-line monitoring tool (WP5).
CICe	**RTD partner**. CICe is a non-profit energy research centre focuses on basic research in energy related materials oriented to storage applications with the aim of contributing to industrial competitiveness through: excellent and breakthrough research; transfer of technology and knowledge to industry.	**Main Role: support to batteries use case.** CICe will mainly participate in WP4 and WP6. In WP4 CICe will participate in the elaboration of the database related to LIPSS process, specifically on the processes associated to metals typically used in batteries. Its main contribution will be in WP6, being involved in the batteries use case.
Cegasa (TP of CICe)	**Industry. (Third Party of CICe)** CEGASA Portable Energy is focused on the production of manganese dioxide to be used in the manufacture of alkaline batteries and on the design and manufacture of energy storage systems based on two different technologies zinc-air and lithium-ion.	CEGASA will be one of the end users that will validate the use of the developed laser texturing prototype, particularly, for the functionalization of current collectors to be incorporated in its batteries. It will also be involved in WP7 to maximise the future impact of the project results on its processes.
FAGOR	**Industry.** FAGOR has a great experience in the development and manufacturing of products for machine automation and control. Its wide technical and sales network spread out in more than 50 countries through its own branch offices and distributors.	**Main Role: use case on machine-tool sector.** FAGOR will be one of the end users that will validate the use of the developed laser texturing prototype, particularly, for the functionalization of optical encoders. It will also be involved in WP7 to maximise the future impact of the project results on its processes.
RESCOLL	**SME** specialized in the manufacturing of implantable materials for different clients in the orthopedic, spinal, and maxillofacial domains	**Main Role: use case on biomedical implants** RESCOLL Group will be one of end-user test case of the LASER4SURF system, particularly on dental implants. It will also be involved in WP7 to maximise the future impact of the project results on its processes.
ESCI	**not-for-profit organisation** that empowers scientists and engineers, helping them to communicate with a non-expert audience to bridge the gap between science, technology and society.	**Main Role: dissemination and communication.** ESCI will lead the communication and dissemination activities in WP8. ESCI will identify the communication needs for the specific audiences and find and use the channels of communication needed to fulfil this task.

In addition, VISUM envisages that part of its work will be performed by IRIS (Innovació i Recerca Industrial i Sostenible, S.L.), linked third party to VISUM.

The involvement of IRIS will allow to increase the level of excellence of the technical work planned as IRIS staff are familiar with validation of the monitoring devices performance and control systems development. Particularly, IRIS will be responsible for the following tasks:

- Task 3.2 Data acquisition and processing software development
- Task 3.3 Functional prototype development (IRIS)

Besides, it will collaborate in the integration of the in-line monitoring in the global system as well as in the dissemination and exploitation tasks.

3.3.2 INDUSTRIAL/COMMERCIAL INVOLVEMENT IN THE PROJECT TO ENSURE EXPLOITATION OF THE RESULTS

LASER4SURF project counts with 5 industrial companies (4 as partners and 1 as Third party) with different roles that will help to maximised the exploitation of the results.

Firstly, LASEA involvement in the project means not only a technical contribution as integrator of the final system, but also a strategical industrial and commercial partner due to its experience on the manufacture and sales of laser production lines all over the world (Europe, US, Australia). This experience and broad international presence will be a key aspect to ensure the future exploitation of the results.

Secondly, VISUM (inline monitoring developer) will help to extend the exploitation possibilities of the solution through its current and future clients.

Finally, the participation of 3 companies as end users (2 as partners and 1 as third party) will be essential for the success of the exploitation plan, as they will represent the first validation of LASER4SURF system in an operational environment. For this reason, they will be involved in the internal exploitation activities in order to consider their experience in the definition of exploitation activities.